Nakamoto Distributed Consensus as a Non-Equilibrium Phase Transition
A Phenomenological Statistical-Physics Description

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Moving beyond traditional cryptographic and game-theoretic analyses, we model Nakamoto consensus as a dissipative structure maintained far from thermodynamic equilibrium. By mapping macroscopic network observables onto a one-dimensional Ginzburg–Landau framework, we propose a falsifiable phenomenological description of blockchain immutability grounded in non-equilibrium statistical physics. In this description, the emergence of a canonical history appears as a continuous ordering transition, and time is treated not as an externally imposed global clock, but as an *emergent coarse-grained variable* driven by continuous thermodynamic dissipation. By coupling network power, latency, and block interval into an effective temperature, the model yields a protocol-dependent upper bound on the informational volume of bytes per block ($V_{\mathrm{B,crit}}$) and anchors the ledger's macroscopic stability in the embodied exergy of the physical hardware substrate. In particular, computational validation latency (γ) imposes an irreducible constraint even in the formal limit of infinite communication bandwidth. We further interpret the Unspent Transaction Output (UTXO) set as an effective Markov blanket providing a compressed state description, and we analyze deterministic subsidy step changes (the protocol's "Halving") as non-equilibrium thermodynamic quenches. Within this framework, the observed dynamics are consistent with a persistent violation of the fluctuation–dissipation theorem, maintained by the continuous expenditure of exogenous exergy.

I. INTRODUCTION: THE THERMODYNAMICS OF DISTRIBUTED CONSENSUS

The essential concern in distributed consensus is the inability to define a canonical order for events in the absence of a global clock. Classical Byzantine Fault Tolerance (BFT) protocols typically rely on authenticated voting and logical ordering rather than an explicit thermodynamic cost; in open-membership settings, such schemes require separate Sybil-resistance assumptions [1, 2]. Physically, the double-spending problem represents a failure of time-ordering invariance: lacking an irreversible thermodynamic cost, $t \to -t$ symmetry makes history $\mathbf{h}_A = \{E_1, E_2\}$ indistinguishable from $\mathbf{h}_B = \{E_2, E_1\}$ [3]. Extracting a unique, immutable history from the stochastic noise of a peer-to-peer network requires a symmetry-breaking process driven by the energy dissipation of its internal structure [4].

Recent advances in information physics have begun mapping traditional thermodynamic ensembles onto distributed protocols, demonstrating that consensus dynamics can be modeled as irreversible processes driven by entropy reduction, where blockchain stability acts as a Lyapunov function [5]. Building upon this emerging consensus-physics isomorphism, we present a minimal phenomenological model of the Nakamoto protocol as a non-equilibrium thermodynamic system. To maintain a strictly objective view of the technology, our analysis intentionally decouples the treatment of financial information from the underlying physical propagation of data blocks.

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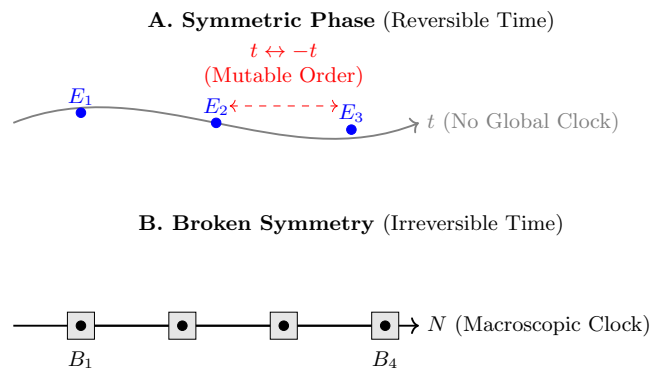

Figure 1: **Arrow of time in the effective description.** Top: without substantial dissipation, event ordering remains effectively time-symmetric and fluid. Bottom: sustained Proof-of-Work dissipation selects a stable macroscopic sequence of blocks.

Accordingly, we treat the chain tip not as a register of economic value, but as a purely informational state transition independent of transaction content.

In our effective description, the distributed ledger is conceptualized as a one-dimensional growing network $\mathcal{L}$ evolving under dissipative forcing. The development of consensus is described as a continuous phase transition between a high-entropy ("disordered") phase, the mempool, and a low-entropy ("ordered") phase, the blockchain, analogous to a liquid-crystallization process [6]. In this view, the blockchain is a dissipative structure whose low entropy is preserved far from equilibrium only through the continuous consumption of work [6, 7].

This manuscript extends a preliminary thermodynamic framework introduced in previous work [8]. While a foundational consensus-physics analogy was previ-

ously outlined, we now map alternative histories to absorbing-state transitions via directed percolation, introduce generalized Langevin dynamics to capture heavy-tailed macroeconomic exergy shocks, establish an effective thermal bound based on computational validation latency, and present empirical observations consistent with fluctuation–dissipation violation during Halving quenches.

The remainder of the paper is organized as follows. Section II establishes the path-weighted statistical framework and emergent thermal time. Section III constructs the Ginzburg–Landau description and derives the block-size bound. Sections IV and V treat finality as correlation decay and the UTXO set as a Markov blanket, respectively. Section VI analyzes transverse stability via the XY model. Section VII models the Halving quench and presents empirical consistency checks. Section VIII summarizes the framework and its limits. Before proceeding, Table I establishes the effective dictionary of correspondences used throughout.

Stat-phys quantity	**Consensus observable**	**Role in the effective description**
Local state variable $(\vec{s}_i, \theta_i)$	Local node view / branch preference	Microscopic or mesoscopic degree of freedom on the peer-to-peer network.
Global order parameter (Φ, ϕ)	Consensus coherence / normalized security density	Macroscopic degree of ordering and symmetry breaking in the coarse-grained ledger state.
Effective temperature (T_{eff})	$\epsilon_0^2 \tau_L / (k_B \bar{E}_{\text{eff}} \tau_B)$	Effective agitation scale induced by latency relative to dissipative ordering.
Mass gap $(\mathfrak{m} = \xi^{-1})$	Reorganization suppression scale	Inverse correlation length governing the depth dependence of reorganization decay.
Dissipation functional $(\Sigma(\mathbf{B}))$	Cumulative proof-of-work expenditure	Coarse-grained path-weighting functional favoring more strongly dissipative histories.
Coupling scale $(\mathcal{E}_{\text{coupling}})$	Latency–security tension	Effective restoring scale opposing local branch misalignment.
Active noise (ζ)	Block-discovery variance / exogenous shocks	Non-equilibrium stochastic forcing driving deviations from equilibrium-like relaxation.
Effective Markov blanket $(\partial\mathcal{V})$	UTXO set	Compressed boundary state mediating validation between ledger history and newly proposed transactions.
External quench	Halving event	Deterministic protocol shock driving transient non-equilibrium response.
Anticipatory drift $(\mathcal{A})$	Pre-Halving hardware deployment	Mobility-weighted forward-looking drift induced by the anticipated subsidy reduction.
Probability density (ρ or P)	Ensemble distribution of coarse-grained consensus states	Macroscopic probability density entering the Fokker–Planck description of order-parameter fluctuations.

Table I: **Phenomenological consensus / statistical-physics dictionary.** The table summarizes the effective correspondences used throughout this phenomenological statistical-physics description of Nakamoto consensus.

Scope, Assumptions, and Limitations

The present framework is intentionally *phenomenological*. Rather than deriving Nakamoto consensus from microscopic first principles, it constructs a coarse-grained, dimensionally consistent description that maps observable protocol variables onto effective thermodynamic quantities.

The analysis is restricted to macroscopic observables—including hashrate, block interval, propagation latency, and reorganization depth—and does not attempt a microscopic account of node-level decision dynamics. It is also specialized to Proof-of-Work systems with Nakamoto-style longest-chain selection and is not intended as a universal theory of consensus.

Several concepts used below, including effective temperature, mass gap, Markov blanket, vortex unbinding, and Maxwell-demon-like filtering, are introduced as effective coarse-grained descriptors rather than literal microscopic identities.

a. Epistemic status of the model. We distinguish three epistemic levels throughout the manuscript: (i) quantities fixed directly by protocol observables or dimensional analysis; (ii) phenomenological fields, coefficients, and constitutive relations; and (iii) analogy-based physical interpretations. The present work claims the greatest robustness at levels (i) and (ii), while statements at level (iii) should be read as heuristic rather than microscopic. Each variable is tagged accordingly in Appendix A (Table I).

Direct quantitative predictions include exponential reorganization-depth suppression, T_{eff} scaling, and a critical block volume $V_{\text{B,crit}}$. Results involving anticipatory dynamics or critical behavior depend on effective couplings not fixed from first principles. The aim is a falsifiable coarse-grained framework for Nakamoto consensus, rather than a complete microscopic theory.

II. NETWORK MODEL, STATISTICS, AND THE ARROW OF TIME

A. The Minimal Network Model and Consensus Hamiltonian

We model the blockchain as a growing one-dimensional network $\mathcal{L}$ of length $N(t)$, where each site i represents a block B_i. The macroscopic state of the system is defined by the consensus-selected block sequence $\mathbf{B}^* = (B_0, B_1, \ldots, B_N)$, corresponding to the heaviest valid chain. Candidate history paths are denoted $\mathcal{C} = (B_0, B_1, \ldots, B_N) \in \Omega$. Unlike static data structures, this network is dynamic; its geometry is determined by a stochastic growth process fighting against entropic degradation (information loss via forks).

In a classic closed thermodynamic system, the ground state emerges through the minimization of internal energy. However, the blockchain is an inherently open dissipative structure, coupled to a global exergy bath.

Motivated by the Maximum Power Principle [9] and related non-equilibrium heuristic ideas [9, 10], we model this open system phenomenologically as one whose long-lived histories are biased toward larger cumulative irreversible dissipation.

Consequently, to extract a unique history, we move away from equilibrium energy minimization and define a macroscopic **dissipative path functional** $\Sigma[\mathcal{C}]$. In this effective description, the ledger is statistically biased toward history trajectories associated with a larger cumulative expenditure of irreversible work. We write

$$\Sigma[\mathcal{C}] = \sum_{i=0}^{N} E_{\text{eff}}(B_i)\, \mathbb{I}_{\text{valid}}(B_i)\,, \tag{1}$$

where $\mathcal{C} = (B_0, B_1, \ldots, B_N)$ denotes a candidate history path of blocks. The functional $\Sigma[\mathcal{C}]$ is extensive in the path length N and carries units of energy [J]. Here $E_{\text{eff}}(B_i)$ is the coarse-grained characteristic energy dissipated in producing block B_i, while $\mathbb{I}_{\text{valid}}(B_i) \in \{0, 1\}$ acts as a Boolean admissibility constraint enforcing protocol validity. The notation $\mathcal{C}$ is reserved throughout for history paths.

We define the state space of the network Ω as the set of coarse-grained ledger histories (including forked histories) accessible over a finite observation window. We then introduce, as a phenomenological ansatz, an effective path weight for a candidate history $\mathcal{C} \in \Omega$:

$$\mathcal{P}[\mathcal{C}] = \frac{1}{\mathcal{Z}_\Sigma} \exp\!\big[\beta_{\text{eff}}\, \Sigma[\mathcal{C}]\big]\,, \tag{2}$$

with

$$\mathcal{Z}_\Sigma = \sum_{\mathcal{C}\in\Omega} \exp\!\big[\beta_{\text{eff}}\, \Sigma[\mathcal{C}]\big]\,. \tag{3}$$

Here $\Sigma[\mathcal{C}]$ denotes the coarse-grained cumulative *irreversible entropy production* associated with candidate history $\mathcal{C}$, so that $\beta_{\text{eff}}\,\Sigma$ is dimensionless. The positive sign in the exponential is intentional: unlike an equilibrium Boltzmann weight, this effective non-equilibrium weighting favors histories sustained by larger irreversible throughput. The consensus-selected chain is then $\mathbf{B}^* = \arg\max_{\mathcal{C}\in\Omega} \Sigma[\mathcal{C}]$, which in the low-temperature limit $\beta_{\text{eff}} \to \infty$ dominates the path measure. We do *not* claim Eq. (2) as a microscopic derivation from first-principles fluctuation theory (such as the Gallavotti–Cohen or Jarzynski relations); rather, it serves as a controlled phenomenological closure for the coarse-grained model, motivated by the generic non-equilibrium result that trajectories with higher entropy production are exponentially more probable [11]. For any finite observation window, the set of coarse-grained accessible histories is effectively finite and the associated cumulative entropy production is bounded above, ensuring the finiteness of $\mathcal{Z}_\Sigma$ within the present coarse-grained description. We emphasize that Eq. (3) is introduced as an *effective non-equilibrium ansatz* for coarse-grained history selection, not as a first-principles microscopic fluctuation theorem, not as a microscopic equilibrium ensemble derived from detailed balance. Its role is to encode, at coarse-grained level, the empirical bias of Nakamoto-style consensus toward histories with larger cumulative irreversible expenditure. In that sense, the construction is closer in spirit to the **maximum caliber** (MaxCal) principle [12, 13], which selects trajectory ensembles by maximizing the path entropy subject to dynamical constraints, than to a canonical partition function of equilibrium statistical mechanics. In the MaxCal framework, Lagrange multipliers conjugate to trajectory observables produce exponential path weights; the positive sign in Eq. (2) enforces a constraint on *expected cumulative dissipation*, providing the formal justification for the non-standard sign convention.

B. Time-Reversal Symmetry and Sybil Degeneracy

Consider a distributed ledger system with zero energetic cost for block creation ($E_{\text{eff}} = 0$). In this regime, the system exhibits **Time-Reversal Symmetry** ($\mathcal{T}$-Symmetry). For any candidate history $\mathcal{C}_A$ (e.g., "E_2 precedes E_3" in Fig. 1), there exists a computationally symmetric candidate history $\mathcal{C}_B$ (e.g., "E_3 precedes E_2") that is indistinguishable to a new observer [3]. This degeneracy implies that the "arrow of time" is undefined; the system exists in a superposition of microstates with equal statistical weight.

Mathematically, this corresponds to a system at infinite temperature ($T \to \infty$, $\beta_{\text{eff}} \to 0$). The partition function diverges, and the probability of observing a specific candidate history $\mathcal{C}$ becomes uniform:

$$\mathcal{P}[\mathcal{C}_A] \approx \mathcal{P}[\mathcal{C}_B] \approx \frac{1}{|\Omega|}\,, \tag{4}$$

where $|\Omega|$ is the cardinality of the discrete state space (all possible forks). In this thermodynamic context, a "Sybil" attack corresponds to a high-degeneracy regime in which many alternative candidate histories carry comparable statistical weight—a thermodynamic degeneracy where costless identities can generate arbitrary histories [2]. Without a symmetry-breaking field, the history entropy S_{history} is maximized, no consensus chain $\mathbf{B}^*$ can be distinguished, and the ledger is rendered purely random and devoid of information [7].

C. Microscopic Substrate and the Nakamoto Action

To ground the macroscopic state in a fundamental physical substrate, we must define the network's fundamental quantum of action before constructing its temperature. We isolate the **Nakamoto Action** h_N (J · s), defined kinematically at the ASIC hardware level:

$$h_N(t) = e_{\text{hash}}(t) \cdot \tau_{\text{hash}}\,, \tag{5}$$

where $e_{\rm hash}(t)$ is absolute physical energy per hash [J] and $\tau_{\rm hash}$ is the execution time of a single SHA-256 state transition [s]. This strips away arbitrary denominations, ensuring the product cleanly yields physical action dimensions.

Unlike Planck's constant h, $h_N(t)$ is a non-stationary, dissipative technological parameter. It dynamically revises downward with semiconductor efficiency, asymptoting toward Landauer's limit.

From this micro-canonical action, we extract the absolute minimum energy threshold required to perform a thermodynamic operation in the network, ϵ_0:

$$\epsilon_0 \equiv e_{\rm hash}(t) = \frac{h_N(t)}{\tau_{\rm hash}} \,. \tag{6}$$

This parameter ϵ_0 acts as the fundamental, microscopic coupling constant that bridges hardware physics to macroscopic consensus stability.

A rigorous timescale separation is imperative. $h_N(t)$ operates at the *micro-canonical* substrate scale ($t \sim \tau_{\rm hash} \sim 10^{-11}$ s), acting as the absolute phase-space floor enforcing Landauer's limit for the logic-gate operations of a SHA-256 transition. Conversely, the ledger's *macro-canonical* dynamics operate on the scale of $\tau_B \approx 600$ s. The macroscopic state energy $\bar{E}_{\rm eff}$ emerges from the statistical integration of billions of micro-canonical fluctuations (the Poisson process). This coarse-graining makes consensus robust and mathematically agnostic to the specific hardware architecture, provided the system respects the physical anchor of $h_N(t)$ and its extracted energy scale ϵ_0.

D. Information Temperature and Thermodynamic Regimes

Unlike equilibrium models governed by an external heat bath, the ledger's effective temperature $T_{\rm eff}$ is an *endogenous* quantity: it measures the stochastic agitation caused by spatial desynchronization across the peer-to-peer graph. Propagation-delay uncertainty acts as thermal noise resisting alignment of the consensus field.

a. Block energy. Each block B_i carries an effective energy cost

$$E_{\rm eff}(B_i) \;\equiv\; W_{\rm PoW}(B_i) + k_B T_{\rm env}\, \mathcal{I}_{\rm val}(B_i)\,, \tag{7}$$

where $W_{\rm PoW}(B_i)$ [J] is the mean proof-of-work energy expended, and $\mathcal{I}_{\rm val}(B_i)$ [bit] is the information-theoretic validation cost converted to energy via the Landauer factor $k_B T_{\rm env} \ln 2$ per bit (with $\ln 2$ absorbed into $\mathcal{I}_{\rm val}$ for compactness). The ensemble average $\bar{E}_{\rm eff} \equiv \langle E_{\rm eff}(B_i)\rangle$ represents the characteristic energy scale set by the protocol's difficulty target.

b. Effective temperature. The construction of $T_{\rm eff}$ must satisfy two constraints: (i) $\beta_{\rm eff} = (k_B T_{\rm eff})^{-1}$ has dimensions J^{-1}, so that the path-weight exponent $\beta_{\rm eff} \sum_i E_{\rm eff}(B_i)$ is dimensionless; and (ii) the Sybil limit $\bar{E}_{\rm eff} \to 0$ maps to $T_{\rm eff} \to \infty$ (maximal disorder). We introduce a microscopic energy quantum ϵ_0 [J], representing the minimal single-hash energy cost (as previously derived in Sec. II C), and define

$$k_B T_{\rm eff} \;\equiv\; \frac{\epsilon_0^2}{\bar{E}_{\rm eff}} \left(\frac{\tau_L}{\tau_B}\right)\,, \tag{8}$$

with inverse

$$\beta_{\rm eff} \;\equiv\; \frac{1}{k_B T_{\rm eff}} \;=\; \frac{\bar{E}_{\rm eff}}{\epsilon_0^2} \left(\frac{\tau_B}{\tau_L}\right)\,. \tag{9}$$

Here τ_B is the mean block interval (work-induced crystallization time) and τ_L the characteristic network latency (spatial uncertainty diffusion time). Equation (8) isolates the competition between these two timescales: when latency dominates ($\tau_L \gg \tau_B$), thermal noise is large; when proof-of-work dominates ($\tau_B \gg \tau_L$), the system cools. Note that $\bar{E}_{\rm eff}$ sets the macroscopic thermal scale, while the per-block stochastic quantity $E_{\rm eff}(B_i)$ enters the path-weight ansatz (2); these are conceptually distinct.

c. Phase regimes. The thermodynamic phases of the ledger follow directly:

- **High temperature** ($\beta_{\rm eff} \to 0$, $\tau_L \gg \tau_B$): Spatial diffusion entropy dominates. Forks proliferate and history is mutable—the "liquid" regime.
- **Low temperature** ($\beta_{\rm eff} \to \infty$, $\tau_L \ll \tau_B$): Negentropy flux dominates. The chain freezes into a rigid, immutable history—the "crystalline" regime.

The critical temperature T_c is set by the critical latency ratio Γ_c (defined in Sec. III A): from Eq. (8), $k_B T_c = \epsilon_0^2/(\bar{E}_{\rm eff}\,\Gamma_c)$.

d. Effective free energy. We define the effective free energy

$$\mathcal{F} \;=\; -\langle \Sigma \rangle + T_{\rm eff}\, S_{\rm th}\,, \tag{10}$$

where $\langle \Sigma \rangle$ [J] is the expected cumulative dissipation and $S_{\rm th}$ [$\mathrm{J\,K^{-1}}$] the thermodynamic entropy of the history ensemble (not the dimensionless Shannon entropy). The sign convention mirrors standard Landau theory with Σ playing the role of a binding energy: maximizing Σ at fixed $T_{\rm eff}$ is equivalent to minimizing $\mathcal{F}$, so the positive-sign path measure (2) selects free-energy minima. In the quasi-stationary regime where $T_{\rm eff}$ varies slowly, $\mathcal{F}$ serves as an approximate Lyapunov functional ($d\mathcal{F}/dt \le 0$) for relaxation toward the dominant history [5]. Far from the adiabatic limit, transient free-energy increases may occur during quench-like events (e.g., sudden hashrate drops).

Nakamoto consensus is thus modeled as a driven ordering process: the longest-chain rule selects candidate histories $\mathcal{C}$ with larger cumulative dissipation—i.e., lower effective free energy. A "51% attack" is recast as a thermally induced melting transition in which the localized effective temperature of the attack vector exceeds the critical binding temperature T_c.

E. Spontaneous Symmetry Breaking (SSB)

The transition from the mempool to the blockchain corresponds to **Spontaneous Symmetry Breaking (SSB)**. We define the Global Order Parameter Φ as the **net magnetization** of the network, by analogy with the Ising model:

$$\Phi \equiv \left| \frac{1}{\mathcal{N}_{\text{nodes}}} \sum_{i=1}^{\mathcal{N}_{\text{nodes}}} \langle \vec{s}_i \rangle \right| , \tag{11}$$

where $\vec{s}_i$ is the "consensus spin" vector of node i.

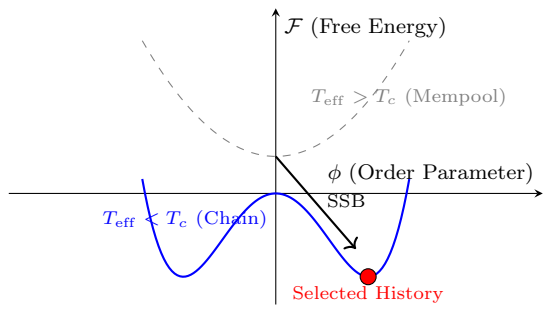

Figure 2: **Symmetry breaking in consensus.** At high effective temperature ($T_{\text{eff}} > T_c$), the local free-energy density $f(\phi)$ has a single minimum at $\phi = 0$ (disorder / Sybil). Below the critical temperature ($T_{\text{eff}} < T_c$), the potential bifurcates into a double-well structure. The system must spontaneously "roll" into one of the stable vacua $\pm\phi_0$ (a specific history), thereby breaking the $\mathbb{Z}_2$ symmetry. Returning to the other vacuum (double spending) requires overcoming the potential barrier $\Delta\mathcal{F}$, which grows with confirmation depth. Here ϕ denotes the local order parameter; the global average $\Phi = \langle \phi \rangle_x$ is introduced in Sec. VII B.

- **Disordered Phase ($T_{\text{eff}} > T_c$):** $\Phi \approx 0$. Nodes are randomly oriented (forks). At a coarse-grained level, the system may be treated as approximately rotationally symmetric in its branch-selection space, with no preferred history.
- **Ordered Phase ($T_{\text{eff}} < T_c$):** $\Phi \to 1$. Nodes spontaneously align along a single "worldline." Rotational symmetry is broken.

This phase transition is induced by the injection of low-entropy energy (electricity), which acts as an effective symmetry-breaking field aligning the spins.

F. Emergence of Thermal Time and Fluctuation Theorem

From the two fundamental timescales we define the dimensionless *latency-control ratio*

$$\Gamma \equiv \frac{\tau_B}{\tau_L} , \tag{12}$$

comparing the protocol block interval to the coarse-grained network-latency timescale. Large Γ (fast propagation relative to block production) favours consensus; small Γ signals the approach to the disordered phase.

In asynchronous distributed networks, there is no fundamental global clock shared by all nodes. The ordering variable relevant for consensus must therefore arise from the system's own irreversible dynamics rather than being imposed externally. This perspective is naturally motivated by the **Thermal Time Hypothesis** [14], according to which the physically relevant notion of time flow is generated by the statistical state of the system itself. In the present work, however, we do not attempt a literal derivation of modular flow. Instead, we use the thermal-time viewpoint as a phenomenological guide for constructing a coarse-grained internal clock tied to irreversible consensus evolution.

For Nakamoto consensus, the natural microscopic obstruction to perfectly synchronous ordering is the network propagation latency τ_L. This latency sets the timescale over which competing local views can coexist before being dissipatively resolved by Proof-of-Work. We therefore define the dimensionless thermal-time increment operationally by

$$d\tau_{\text{th}} \equiv \frac{dt}{\tau_L} , \tag{13}$$

so that thermal time counts the number of latency windows elapsed during the network's irreversible evolution. In this sense, τ_{th} is not an externally imposed coordinate time, but a coarse-grained internal clock measuring the opportunity for consensus reconfiguration.

This kinematic definition is consistent with the energetic structure of the model. Using the effective-temperature relation introduced in Sec. III A,

$$k_B T_{\text{eff}} = \frac{\epsilon_0^2 \tau_L}{\bar{E}_{\text{eff}} \tau_B} = \frac{\epsilon_0^2}{\bar{E}_{\text{eff}}} \Gamma^{-1}, \qquad \Gamma \equiv \frac{\tau_B}{\tau_L} \text{ [Eq. (12)]}, \tag{14}$$

the same latency ratio Γ that governs the thermal-time flow also sets the effective thermal scale. By defining $\mathcal{E}_{\text{shot}} \equiv \epsilon_0^2 / \bar{E}_{\text{eff}}$ as the characteristic energy scale of the Poissonian discovery process, we obtain $k_B T_{\text{eff}} = \mathcal{E}_{\text{shot}} \Gamma^{-1}$, directly linking macroscopic agitation to the microscopic "shot noise" of mining.

Integrating Eq. (13) over one target crystallization epoch τ_B gives

$$\Delta\tau_{\text{th}} = \int_0^{\tau_B} \frac{dt}{\tau_L} = \frac{\tau_B}{\tau_L} \equiv \Gamma , \tag{15}$$

so that one target block-formation interval corresponds to a dimensionless thermal-time increment set by the latency-control ratio. Accordingly, Γ is best interpreted not as a literal quantum of time, but as the natural coarse-grained thermal-time increment associated with one crystallization cycle.

Because the Difficulty Adjustment Algorithm (DAA) continuously regulates the target block interval, the macroscopic ledger clock remains locked, on average, to the protocol timescale:

$$\frac{dN}{dt} \approx \frac{1}{\tau_B} , \tag{16}$$

where N is block height. In this effective sense, the DAA plays a thermostat-like role: it does not fix a literal microscopic temperature, but stabilizes the characteristic crystallization timescale relative to external physical time.

We next quantify the irreversibility of this emergent time—and hence the suppression of reverse ledger evolution—using the **Crooks Fluctuation Theorem** [15] as an effective non-equilibrium analogy. For a coarse-grained ledger transition from state A to state B, we write

$$\ln \frac{\mathcal{P}_F(W)}{\mathcal{P}_R(-W)} = \beta_{\text{eff}} \left(W - \Delta\mathcal{F}\right) = \frac{\Delta S_{\text{irrev}}}{k_B}\,, \tag{17}$$

where $\mathcal{P}_F$ denotes the probability of the forward (honest-validation) trajectory, $\mathcal{P}_R$ the probability of the corresponding reverse (reorganization) trajectory, W the total coarse-grained dissipative work, and $\Delta\mathcal{F}$ an effective free-energy-like barrier between the two macrostates. Equation (17) should be read as a phenomenological path-weighting relation for the irreversible consensus dynamics, rather than as a claim that the blockchain is a near-equilibrium thermal system in the strict microscopic sense.

The corresponding macroscopic entropy-production rate is modeled as

$$\dot{S}_{\text{irrev}}(t) = \frac{1}{T_{\text{eff}}(t)} \left(\frac{\bar{E}_{\text{eff}}}{\tau_B} - \frac{d\mathcal{F}}{dt} \right) \geq 0\,, \tag{18}$$

where $\bar{E}_{\text{eff}}/\tau_B$ is the average macroscopic dissipative power required to sustain block formation. For the relaxational sign convention adopted in Eq. (10), the ordered regime satisfies $d\mathcal{F}/dt < 0$, so the bracketed term is positive and hence $\dot{S}_{\text{irrev}} > 0$ provided $T_{\text{eff}}(t) > 0$. Equation (18) should be interpreted as an effective coarse-grained entropy-production balance rather than a microscopic thermodynamic identity. In this sense, immutability emerges as a statistical bias generated by persistent dissipation.

The network therefore does not merely record history; at the coarse-grained level it suppresses alternative histories by sustaining a strictly positive rate of irreversible entropy production. Per the Landauer–Bennett bound, each irreversibly erased bit of discarded alternative-chain state contributes at least $k_B T_{\text{sub}} \ln 2$ of dissipated heat. Dividing the associated minimal heat-dissipation rate by T_{sub} gives the bound

$$\dot{S}_{\text{irrev}} \;\geq\; k_B \ln 2 \, \frac{dN_{\text{bit}}^{(\text{erased})}}{dt} \;>\; 0\,, \tag{19}$$

where $N_{\text{bit}}^{(\text{erased})}$ counts the cumulative number of orphaned-block and rejected-state bits discarded by the network. This persistent dissipation selects a macroscopically time-asymmetric ledger trajectory and provides the thermodynamic basis, within the present effective description, for the suppression of history reorganizations.

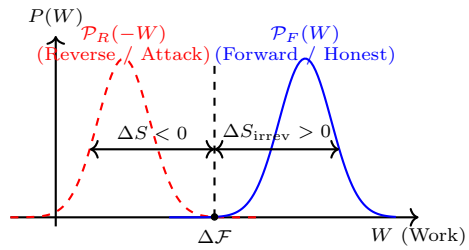

Figure 3: **Effective Crooks asymmetry in Nakamoto consensus.** Schematic forward and reverse work distributions for honest extension and adversarial reorganization within the coarse-grained effective description. Their intersection occurs at $\Delta\mathcal{F}$. For a deeply confirmed block, the operating point lies far in the forward tail ($W \gg \Delta\mathcal{F}$), so reverse trajectories are exponentially suppressed in the same phenomenological non-equilibrium ensemble.

III. THERMODYNAMIC POTENTIAL AND SYMMETRY BREAKING

To describe the ledger's macroscopic stability, we construct a one-dimensional coarse-grained Ginzburg–Landau theory, representing the blockchain by a scalar field $\phi(x,t)$ that denotes the normalized local security density along the chain. However, before formally writing the continuous free-energy functional (Sec. III B), we must first map the discrete, physical kinematics of the P2P network into a continuous thermal control parameter.

A. Effective Temperature, Microscopic Latency, and the Block-Size Limit

The chain "tip" behaves as a fluid interface subject to thermal roughness (orphan blocks). We define the Effective Temperature T_{eff} as the energy dissipated "blindly" due to spatial desynchronization. This T_{eff} measures the consensus graph's stochastic agitation and must be distinguished from the substrate's true thermal temperature T_{sub}. While T_{sub} dictates Landauer's limit per hash, T_{eff} governs macroscopic consensus stability against latency noise.

Since the Boltzmann-like parameter is $\beta_{\text{eff}} = (k_B T_{\text{eff}})^{-1}$, we obtain $\beta_{\text{eff}} = \Gamma \frac{\bar{E}_{\text{eff}}}{\epsilon_0^2}$, which yields the explicit relation

$$T_{\text{eff}} = \frac{\epsilon_0^2}{k_B \bar{E}_{\text{eff}}\, \Gamma} = \frac{\epsilon_0^2\, \tau_L(V_B)}{k_B \bar{E}_{\text{eff}}\, \tau_B}\,, \tag{20}$$

showing that T_{eff} increases linearly with network latency τ_L and decreases with block interval τ_B, consistent with the qualitative prediction of Sec. VII D (item 2). Here T_{eff} should be understood as a coarse-grained agitation scale entering the phenomenological theory in the same algebraic role as temperature in standard ordering problems; it is not an equilibrium thermodynamic temperature of the substrate. However, network latency $\tau_L(V_B)$ is a kinetic emergent property of the P2P graph topology.

Extending models by Croman *et al.* [16] and Gervais *et al.* [17], we integrate the actions required for a block of volume V_B to traverse a network of diameter $D_{\rm hop}$ (average hops to global saturation). A single-hop propagation requires geometric transmission ($\tau_{\rm link}$), bandwidth-constrained transfer, and computational validation. The single-hop time $\tau_{\rm hop}$ is

$$\tau_{\rm hop}(V_B) = \tau_{\rm link} + V_B\left(\frac{1}{C_{\rm node}} + \gamma\right), \tag{21}$$

where $\tau_{\rm link} \approx d/c$ is the geometric propagation delay, $C_{\rm node}$ is the bottleneck link capacity [Bytes/s], and γ is the validation-time coefficient [s/Byte]. Crucially, γ denotes computational friction, representing the irreducible CPU work required to verify cryptographic signatures and UTXO transitions [16]. Summing over $D_{\rm hop}$ hops to global saturation gives the total network latency:

$$\tau_L(V_B) = D_{\rm hop}\cdot\tau_{\rm hop}(V_B) = D_{\rm hop}\left[\tau_{\rm link} + V_B\left(\frac{1}{C_{\rm node}} + \gamma\right)\right], \tag{22}$$

which is linear in V_B. Substituting into Eq. (12) and setting $\Gamma = 1$ (critical condition $\tau_B = \tau_L$). Setting $a(V_B) = 0$ in Eq. (33) and solving for V_B yields the critical block volume

$$V_{\rm B,crit} = \frac{\tau_B/\Gamma_c - D_{\rm hop}\,\tau_{\rm link}}{D_{\rm hop}\big(C_{\rm node}^{-1} + \gamma\big)}, \tag{23}$$

where $\tau_0 \equiv D_{\rm hop}\,\tau_{\rm link}$ is the aggregate bare propagation delay and Γ_c the critical value of the latency-control ratio $\Gamma \equiv \tau_B/\tau_L$ at which the Landau coefficient a changes sign (i.e. $a(\Gamma_c) = 0$). A physically admissible positive threshold requires $\tau_B/\Gamma_c > D_{\rm hop}\,\tau_{\rm link}$. For $V_B > V_{\rm B,crit}$, the effective Landau coefficient satisfies $a > 0$ in the present sign convention, corresponding to the disordered phase of the phenomenological theory. Here $D_{\rm hop}$ is the relay-hop count (dimensionless), $\tau_{\rm link}$ [s] the *per-hop* link latency, $C_{\rm node}$ [byte s^{-1}] the per-node throughput, and γ [s byte^{-1}] the marginal validation friction. With $\tau_0 \equiv D_{\rm hop}\tau_{\rm link}$ interpreted as the aggregate bare propagation delay, these assignments ensure that $V_{\rm B,crit}$ carries units of [byte].

The macroscopic network latency integrates this hop time across the graph diameter:

$$\tau_L(V_B) \approx D_{\rm hop}\,\tau_{\rm hop}(V_B) = \tau_0 + V_B D_{\rm hop}\left(\frac{1}{C_{\rm node}} + \gamma\right), \tag{24}$$

where $\tau_0 \equiv D_{\rm hop}\cdot\tau_{\rm link}$ is the incompressible geometric delay. From this, we extract the **Effective Global Channel Capacity** $C_{\rm eff}$:

$$C_{\rm eff} = \left[D_{\rm hop}\left(\frac{1}{C_{\rm node}} + \gamma\right)\right]^{-1}. \tag{25}$$

Substituting Eq. (24) into the definition of $\beta_{\rm eff}$ yields

$$\beta_{\rm eff}(V_B) \;= \frac{\bar{E}_{\rm eff}}{\epsilon_0^2}\;\left(\frac{\tau_B}{\tau_0 + V_B/C_{\rm eff}}\right). \tag{26}$$

Increasing block size is physically equivalent to heating the system. Spatial Shannon entropy increases, mechanically raising $T_{\rm eff}$. Thus, a critical block size $V_{\rm B,crit}$ exists where $T_{\rm eff}$ exceeds the phase-transition threshold ($T_{\rm eff} > T_c$). Setting $\Gamma = \Gamma_c$:

$$V_{\rm B,crit} \approx C_{\rm eff}\left(\frac{\tau_B}{\Gamma_c} - \tau_0\right). \tag{27}$$

To anchor this limit, we perform a numerical estimation. Setting a security threshold where the correlation length ξ begins to collapse. Empirically, an orphan rate $r_c \approx 10\%$ corresponds to the regime where approximately one in every Γ_c blocks is wasted due to latency-induced competition. In the simplest mean-field estimate, $r_c \approx \tau_L/\tau_B = \Gamma^{-1}$, which gives $\Gamma_c \approx 1/r_c \approx 10$ at $r_c = 10\%$ [18]. With $\tau_B = 600\,$s and base latency $\tau_0 \equiv D_{\rm hop}\,\tau_{\rm link} \approx 0.5\,$s (the propagation floor from Eq. (22) at zero block volume), the critical size depends solely on the effective throughput $C_{\rm eff}$. Setting $C_{\rm eff} \approx 10\,$MB/s:

$$V_{\rm B,crit} \approx 10\,{\rm MB/s} \times \left(\frac{600\,{\rm s}}{10} - 0.5\,{\rm s}\right) \approx 595\,{\rm MB}\,. \tag{28}$$

To assess the sensitivity of this estimate, we propagate uncertainties in the input parameters. Taking $C_{\rm eff} \in [5, 20]\,$MB/s (reflecting heterogeneous global infrastructure), $\Gamma_c \in [8, 15]$ (corresponding to orphan rates between $\sim 7\%$ and $\sim 12\%$), and $\tau_0 \in [0.3, 1.0]\,$s, the critical block size spans $V_{\rm B,crit} \in [195, 1495]\,$MB. The central estimate of $\sim 600\,$MB should therefore be understood as an order-of-magnitude prediction whose precise value depends on network topology and infrastructure quality. For current Nakamoto-protocol parameters ($V_B = 1{-}4\,$MB), the system operates deep within the ordered regime ($V_B \ll V_{\rm B,crit}$) regardless of the precise bound.

Within this coarse-grained model, the asymptotic limit is controlled by the product $D_{\rm hop}\gamma$, even if communication bandwidth is taken formally large ($C_{\rm node} \to \infty$). Consequently, proposals to increase block sizes to gigabytes would likely push the network beyond its thermal percolation limit. If $V_B \gg V_{\rm B,crit}$, the system abandons the ordered "crystalline" phase for a disordered "liquid" state, fracturing temporal domains.

B. The Order Parameter and Landau Free Energy

Rather than tracking discrete block topology, we analyze the system at the hydrodynamic limit via a mean-field approach. While the underlying P2P topology is a high-dimensional "small-world" graph, we bridge this discrete network to a continuous field theory by coarse-graining highly interconnected clusters onto a one-dimensional spatial manifold x. Here, x represents a parameterized projection along the network's maximum latency path (its topological diameter $L_{\rm net}$). This reduction is justified because macroscopic thermal fluctuations

(forks) propagate linearly along the bottleneck of this effective diameter.

Notational Note on Time Dependence: Nakamoto consensus operates under effective macroscopic timescale separation. Thermal relaxation ($\tau_B \approx 600\,\mathrm{s}$) is infinitesimally shorter than secular hardware evolution. We thus invoke the **adiabatic approximation**: macroscopic variables ($P_{\mathrm{net}}(t)$, $\bar{E}_{\mathrm{eff}}(t)$, $T_{\mathrm{eff}}(t)$) are quasi-stationary when analyzing the potential $\mathcal{F}$. Explicit time dependence (t) is occasionally omitted for brevity.

We use a coarse-grained scalar field $\phi(x)$ to represent the normalized local security density along the chain. We note that in one spatial dimension, the Mermin–Wagner theorem [19] forbids spontaneous breaking of a *continuous* symmetry at finite temperature in equilibrium. This obstruction does not apply here: the local free-energy density $f(\phi)$ [Eq. (30)] possesses a discrete $\mathbb{Z}_2$ (double-well) degeneracy rather than a continuous symmetry, so the theorem's hypotheses are not satisfied. Additionally, the system is driven far from equilibrium by continuous exergy injection, although this second consideration is not required for the $\mathbb{Z}_2$ case.

For the present one-dimensional ledger description, the relevant Ginzburg–Landau functional is

$$\mathcal{F}[\phi] = \int dx \left[\frac{\kappa}{2}\left(\partial_x \phi\right)^2 + f(\phi)\right], \tag{29}$$

where κ is a stiffness coefficient penalizing spatial gradients of the order parameter, and

$$f(\phi) = a(T_{\mathrm{eff}})\,\phi^2 + b\,\phi^4 \tag{30}$$

is the local free-energy density. Since ϕ is dimensionless in our normalization, f, a, and b all carry the dimensions of energy per unit length in this one-dimensional description [$\mathrm{J\,m^{-1}}$], while κ carries the dimensions of energy times length [$\mathrm{J\,m}$]. In the homogeneous consensus state ($\nabla\phi \to 0$), the gradient penalty vanishes, and the system dynamics are governed entirely by this standard "Mexican hat" potential, whose coefficients are determined by network thermodynamic bounds:

1. **Instability Coefficient (a) and Exergetic Susceptibility (χ_E):** The Landau instability parameter a governs symmetry breaking and changes sign at the effective critical temperature T_c:

$$a(T_{\mathrm{eff}}) = \chi_E (T_{\mathrm{eff}} - T_c)\,, \tag{31}$$

where $\chi_E > 0$ is an effective susceptibility converting thermal deviation into free-energy-density variation. In the present one-dimensional coarse-grained description, χ_E has units of $\mathrm{J\,m^{-1}\,K^{-1}}$.

Using $\Gamma = \tau_B/\tau_L$, the quadratic term depends on spatial latency:

$$a(\tau_L) = \mathcal{E}_{\mathrm{coupling}}\left(\Gamma_c\,\frac{\tau_L}{\tau_B} - 1\right), \tag{32}$$

where $\mathcal{E}_{\mathrm{coupling}} \equiv \chi_E T_c$ is an effective coupling free-energy density with units of $\mathrm{J\,m^{-1}}$ in this one-dimensional formulation. Substituting latency $\tau_L(V_B)$ exposes the thermodynamic coupling between validation and symmetry breaking:

$$a(V_B) = \mathcal{E}_{\mathrm{coupling}}\left\{\frac{\Gamma_c}{\tau_B}\left[D_{\mathrm{hop}}\,\tau_{\mathrm{link}} + V_B\,D_{\mathrm{hop}}\big(C_{\mathrm{node}}^{-1} + \gamma\big)\right] - 1\right\}. \tag{33}$$

Here $D_{\mathrm{hop}}\,\tau_{\mathrm{link}}$ is the aggregate bare link latency and $V_B D_{\mathrm{hop}}(C_{\mathrm{node}}^{-1} + \gamma)$ the size-dependent validation delay. Dimensional consistency is maintained by assigning the validation friction γ units of $\mathrm{s\,byte^{-1}}$ (inverse validation throughput). Setting $a(V_B) = 0$ recovers Eq. (27). Consequently, γ enters the effective thermal control parameter as a phenomenological information-processing bottleneck.

   - **Ordered Regime ($a < 0$):** Mining incentives overcome entropic diffusion. The trivial state $|\phi| = 0$ is unstable, and the system settles into consensus $\phi = \pm\phi_0$.
   - **Disordered Regime ($a > 0$):** If friction γV_B or bandwidth limits push T_{eff} past the critical threshold, a becomes positive. The free-energy minimum shifts to the disordered state ($\phi = 0$). Exceeding this validation speed limit drives the system across the order–disorder boundary, destabilizing consensus.

2. **Saturation Term and ASIC Mass ($b > 0$):** The quartic term $b\,|\phi|^4$ provides the restoring force that prevents unbounded hashrate growth, anchoring the field theory in the material constraints of the hardware substrate. Throughout this work the spatially-resolved Landau order parameter $\phi(x)$ is taken to be *dimensionless* (normalised to the equilibrium condensate value), so that all dimensionful content resides in the coefficients a, b, κ, and the integration measure dx [m].

We define **computational mass efficiency** η_M as the ratio of average mining unit mass M_{unit} to raw hashrate H_{unit} ($\approx 1.5 \times 10^{-13}\,\mathrm{kg\cdot s}$ for 2026-era ASIC). Introducing e_{mfg} [$\mathrm{J\,kg^{-1}}$] as the specific **embodied ASIC exergy**—a composite thermodynamic constant representing the total work required to extract, refine, and assemble the active computing infrastructure—we relate the energy barrier b to this physical deployment cost via a dimensionless structural constant c_0. To ensure b properly evaluates to an energy density [$\mathrm{J\,m^{-1}}$] consistent with the 1D Ginzburg–Landau functional, we divide by the effective network spatial diameter L_{net}:

$$b = c_0 \frac{e_{\mathrm{mfg}} \cdot \eta_M \cdot H_{\mathrm{eq}}}{L_{\mathrm{net}}}\,. \tag{34}$$

This dynamically anchors the Landau potential in the material and energetic constraints of the physical substrate.

Order-of-magnitude estimates. While a precise determination of c_0 and $\mathcal{E}_{\text{coupling}}$ requires calibration against orphan-rate data, their orders of magnitude can be bounded from consistency requirements. At the adiabatic equilibrium $|\phi_0| = 1$, the constraint $-a = 2b$ [Eq. (36)] together with Eq. (32) evaluated at $\Gamma = \Gamma_c$ yields

$$\mathcal{E}_{\text{coupling}} \sim 2\, c_0\, \frac{e_{\text{mfg}}\, \eta_M\, H_{\text{eq}}}{L_{\text{net}}}\,. \tag{35}$$

Taking representative 2024 values ($e_{\text{mfg}} \sim 10^8\,\text{J}\,\text{kg}^{-1}$, $\eta_M \sim 1.5 \times 10^{-13}\,\text{kg}\,\text{s}$, $H_{\text{eq}} \sim 6 \times 10^{20}\,\text{s}^{-1}$, $L_{\text{net}} \sim 4 \times 10^7\,\text{m}$) gives $b \approx c_0 \times 2.3 \times 10^8\,\text{J}\,\text{m}^{-1}$. For $c_0 \sim \mathcal{O}(1)$, this places the quartic stabilization scale at the level of hundreds of megajoules per meter of effective network diameter, consistent with the known annual network energy expenditure ($\sim 10^{17}\,\text{J}$) distributed over L_{net} and τ_B. A more precise calibration against empirical orphan-rate measurements as a function of block size would be needed to fix c_0 and $\mathcal{E}_{\text{coupling}}$ independently.

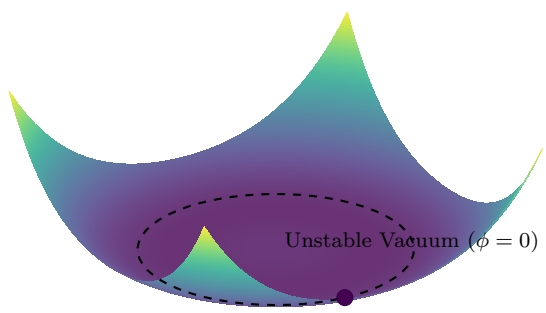

Figure 4: **The consensus potential.** The system spontaneously rolls from the unstable origin (zero security) toward the stable valley ϕ_0 (equilibrium hashrate), defined by the balance between incentive (a) and cost (b).

The system relaxes into a stable ground state with normalized equilibrium hashrate $|\phi_0|$:

$$|\phi_0| = \sqrt{\frac{-a}{2b}}\,. \tag{36}$$

By defining the adiabatic equilibrium state through the normalization $|\phi_0| = 1$ (as used in Sec. VII), we impose the coefficient constraint $-a = 2b$. This should be interpreted as a choice of order-parameter normalization at equilibrium rather than as an independent microscopic identity between the Landau coefficients.

C. Phase-Like Fluctuations and Consensus Rigidity

At a coarse-grained level, local disagreements about the preferred chain tip are represented by phase-like fluctuations of the consensus field, penalized by the stiffness term κ in Eq. (29). This language describes collective synchronization phenomenologically rather than claiming an exact microscopic continuous symmetry.

D. Macroscopic Energy Scale and the DAA Thermostat

To ground the macroscopic energy scale $\bar{E}_{\text{eff}}$ empirically, we define it as the *ensemble average* of the total physical work required to crystallize a single block. This statistical barrier is dictated by the Difficulty parameter $\mathcal{D}$, which imposes an expected hash count $\mathcal{N}_H(\mathcal{D})$. With $e_{\text{hash}}(t)$ representing the absolute physical energy per hash [Joules], the expected block-state energy is

$$\bar{E}_{\text{eff}}(t) = \mathcal{N}_H(\mathcal{D}) \cdot e_{\text{hash}}(t)\,. \tag{37}$$

This statistical barrier dictates an *adiabatic equilibrium hashrate* $H_{\text{eq}}(t)$ required to meet the target block interval τ_B:

$$\bar{E}_{\text{eff}}(t) = e_{\text{hash}}(t) \cdot H_{\text{eq}}(t) \cdot \tau_B\,. \tag{38}$$

Recognizing that the product $e_{\text{hash}}(t) \cdot H_{\text{eq}}(t)$ constitutes the required equilibrium network power $P_{\text{eq}}(t)$ in Watts, the macroscopic scale simplifies to

$$\bar{E}_{\text{eff}}(t) = P_{\text{eq}}(t) \cdot \tau_B\,. \tag{39}$$

Consequently, the macroscopic energy scale $\bar{E}_{\text{eff}}$ is governed by the Difficulty Adjustment Algorithm (DAA), which acts as a non-linear feedback controller. By modulating $\mathcal{N}_H(\mathcal{D})$, the protocol defines the depth of the free-energy well to maintain the canonical $\tau_B \approx 600\,\text{s}$.

However, during a thermodynamic quench (such as a Halving), the actual instantaneous power $P_{\text{net}}(t)$ decouples from the DAA-mandated $P_{\text{eq}}(t)$. To correctly capture the network's transient thermal response out of equilibrium, we substitute the actual instantaneous power $P_{\text{net}}(t)$ into the temperature definition [Eq. (8)], yielding the operational identity:

$$k_B T_{\text{eff}}(t) = \frac{\epsilon_0^2}{P_{\text{net}}(t) \cdot \tau_B} \left(\frac{\tau_L}{\tau_B}\right)\,. \tag{40}$$

which formally demonstrates that dropping the network power $P_{\text{net}}(t) \to 0$ causes the systemic temperature to appropriately diverge, rendering the chain entirely fluid and vulnerable.

Furthermore, the relationship between block-discovery precision and energy dissipation is constrained, at least heuristically, by the **Thermodynamic Uncertainty Relation (TUR)** from modern stochastic thermodynamics [20, 21]. In its integrated hitting-time form, the TUR states that the precision of a stochastic current over an observation window requires a minimum total entropy production σ_{obs}:

$$\frac{\text{Var}(\tau_B)}{\langle \tau_B \rangle^2} \gtrsim \frac{2k_B}{\sigma_{\text{obs}}}\,, \tag{41}$$

where $\sigma_{\rm obs}$ denotes the total entropy production accumulated over the observation interval, carrying units of $\mathrm{J\,K^{-1}}$. In the quasi-stationary regime, the macroscopic entropy produced over a single crystallization epoch is bounded by the ratio of the dissipated block energy to the effective temperature:

$$\sigma_{\rm obs} \approx \frac{\bar{E}_{\rm eff}}{T_{\rm eff}} = k_B \left(\frac{\bar{E}_{\rm eff}}{\epsilon_0}\right)^2 \frac{\tau_B}{\tau_L}\,. \tag{42}$$

Substituting this back into Eq. (41) yields a direct kinematic constraint: the relative variance in block discovery time is fundamentally limited by the spatial latency ratio τ_L/τ_B.

To maintain a predictable block-production timescale (τ_B) against active noise, the protocol requires sustained dissipation. Unregulated hashrate $H(t)$ growth would collapse crystallization time ($\tau_B \to 0$), destroying precision bounds. The DAA thus acts as a physical thermostat, raising $\mathcal{N}_H(\mathcal{D})$ to anchor τ_B at 600 s. By increasing the expected work required per block, the protocol counteracts temperature spikes in the effective description and helps preserve the ordered ledger state.

E. The Consensus Action and Cumulative Barrier

To formalize the accumulation of work over time, we define the **consensus action** over a time window $[t_1, t_2]$ as:

$$\mathcal{S}_{\rm PoW}(t_1,t_2) \;\equiv\; \tau_B \int_{t_1}^{t_2} P_{\rm net}(t)\,dt \;=\; \tau_B\, E_{\rm total}(t_1,t_2)\,, \tag{43}$$

which has the dimensions of action ($\mathrm{J\cdot s}$), since $E_{\rm total} = \int P_{\rm net}\,dt$ is the total energy dissipated over the window. The prefactor τ_B converts accumulated energy into an action-dimensioned quantity by weighting each dissipated Joule by the protocol's characteristic crystallization time. We emphasize that $\mathcal{S}_{\rm PoW}$ is not a variational action in the Lagrangian sense; it is a cumulative dissipation measure that happens to carry action dimensions. In this sense, the ledger accumulates a growing *thermodynamic barrier*, not merely data. By dissipating heat, the protocol decreases logical entropy. In this effective description, the heaviest chain corresponds to the history maximizing the accumulated quantity $\mathcal{S}_{\rm PoW}$, thereby creating a large thermodynamic barrier against reorganization.

IV. CORRELATION LENGTH AND PROBABILISTIC FINALITY

In this section, we derive the probabilistic finality of the ledger not as a geometric horizon but as a correlation-decay property of a massive 1+0-dimensional field theory. Treating the blockchain as a 1D effective spin chain at the finite temperature $T_{\rm eff}$ defined in Sec. IV (see also the Landau coefficients of Sec. III), we show that the mass gap $\mathfrak{m}$ is fixed entirely by the honest-to-adversarial hashrate ratio and that the conventional "six-block rule" emerges as a special case of the exponential decay $e^{-z\mathfrak{m}}$.

A. The Correlation Function

We define ledger stability via the two-point correlation function $G(z)$. Let s_i be the macroscopic block state at height i (confirmed vs. rejected). The probability that the state at depth z remains correlated with the current consensus tip (height N) is given by

$$G(z) = \langle s_N \cdot s_{N-z} \rangle\,. \tag{44}$$

In a massless system (zero difficulty), fluctuations would propagate infinitely, and history would be mutable at all depths ($G(z) \sim$ const). However, the Proof-of-Work mechanism introduces an effective "mass gap" $\mathfrak{m}$ (set by the difficulty target) into the theory. In massive field theories, correlations decay exponentially with distance:

$$G(z) \sim e^{-z/\xi}\,, \tag{45}$$

where ξ is the **Correlation Length**. Because $G(z)$ measures the influence of the consensus tip on a historical block, a successful reorganization represents a total decorrelation event where the historical state $s^{(z)}_{N-z}$ is forcibly flipped by an attacker. We therefore use the decay of $G(z)$ as a proxy for probabilistic finality: as the correlation between the current tip and a block at depth z decreases, the probability of a successful reorganization is correspondingly suppressed. This length scale is determined by the ratio between the "binding energy" (honest hashrate H_H) and the "thermal noise" (attacker hashrate H_A).

B. Derivation of the Mass Gap (The "6-Block" Rule)

By identifying the reorganization probability with the Gambler's Ruin absorption probability [1], $\mathcal{P}_{\rm reorg}$ takes the form of an exponential decay in block depth:

$$\mathcal{P}_{\rm reorg}(z) = \left(\frac{q}{p}\right)^z = e^{-z\cdot\ln(p/q)}\,, \tag{46}$$

where p is the honest network hashrate share and q the attacker share ($p+q=1$). Assuming the system operates in the ordered phase ($p > q$), we compare this to the standard exponential decay in massive field theories ($e^{-z/\xi}$) to extract the inverse correlation length (the mass gap) $\mathfrak{m}$:

$$\mathfrak{m} = \xi^{-1} = \ln\!\left(\frac{p}{q}\right). \tag{47}$$

This mass gap $\mathfrak{m}$ does not represent a fundamental particle mass, but rather an emergent inverse correlation scale arising from the statistical distribution of the network's hashrate. Here $\xi = 1/\mathfrak{m}$ is measured in units of *block depth* (dimensionless, though not restricted to integer values), so that $\mathfrak{m} = \ln(p/q)$ is likewise dimensionless (inverse block-depths). At the critical point $p = q = \frac{1}{2}$, $\mathfrak{m} \to 0$ and $\xi \to \infty$, signaling the divergence of the reorganization correlation length and the breakdown of consensus — the blockchain analog of a massless (gapless) critical mode. For the commonly assumed honest majority $p = 0.9$, $q = 0.1$, we obtain $\mathfrak{m} = \ln(9) \approx 2.20$, yielding $\xi \approx 0.45$ blocks. Within this phenomenological exponential-depth ansatz, the conventional "6-block" confirmation rule ($z = 6$) gives $\mathcal{P}_{\text{reorg}}(6) = e^{-6\times 2.20} \approx 1.7\times 10^{-6}$, of the same order as commonly invoked heuristic security thresholds in the Bitcoin literature [1].

We stress, however, that this is not Nakamoto's exact attack-probability formula. Rather, it is an effective large-depth asymptotic decay law of the form $\mathcal{P}_{\text{reorg}}(\ell) \sim A_{\text{reorg}} e^{-\mathfrak{m}\ell}$, motivated by the present mass-gap analogy, with a nonuniversal prefactor $A_{\text{reorg}} = O(1)$ suppressed in the rough numerical estimate below.

For the more conservative scenario where the attacking hashrate is $q = 0.3$ and the honest network maintains $p = 0.7$, we compute the effective suppression exponent as $\mathfrak{m} = \ln(p/q) = \ln(7/3) \approx 0.847$ (dimensionless). Under the proposed exponential decay law, this yields $\mathcal{P}_{\text{reorg}}(6) \approx \exp(-0.847 \times 6) \approx 6 \times 10^{-3}$. We emphasize that this probability represents the *stochastic likelihood* of a 6-block reversal under these hashrate conditions, conditioned on the exponential ansatz. It does not incorporate game-theoretic factors (cost of attack, reward for reversal, reputation loss), which are separately addressed in mining-economics literature. The empirical corroboration (Sec. VII C) tests whether this exponential ansatz, combined with effective-temperature scaling, captures the observed variance structure of inter-block times around Halving events.

This is consistent with the generic exponential suppression of large deviations in Markov chains [22]: the fork probability decays as $e^{-\mathfrak{m}\, z}$, with $\mathfrak{m}$ playing the role of a large-deviation rate function evaluated at the honest-majority operating point.

Physical Interpretation:

- **Massless Phase ($\mathfrak{m} \to 0$):** As the attacker approaches parity ($p \approx q \to 0.5$), the mass gap $\mathfrak{m} = \ln(p/q) \to 0$ and the correlation length $\xi = a_0/\mathfrak{m} \to \infty$ (in units of blocks). Finality is never reached; the system resides in a "critical state" of infinite susceptibility to reorganization, $\chi_{\text{reorg}} \propto \xi \to \infty$.

- **Massive Phase ($\mathfrak{m} > 0$):** If $p \gg q$, the correlation length is short and history "freezes" rapidly. Introducing the discrete lattice spacing $a_0 = 1$ block, the dimensionless mass gap $\mathfrak{m} = \ln(p/q)$ defines the correlation length via

$$\xi = \frac{a_0}{\mathfrak{m}} = \frac{1\,\text{block}}{\ln(p/q)}, \tag{48}$$

which correctly carries units of blocks. For $(p, q) = (0.9, 0.1)$, $\mathfrak{m} = \ln 9 \approx 2.20$, $\xi \approx 0.46$ blocks, and $P_{\text{reorg}}(6) \approx e^{-13.2} \approx 1.8 \times 10^{-6}$, yielding a value of the same order as the canonical 6-block heuristic security threshold. A stronger adversary $q = 0.38$ gives $\xi \approx 2.04$ blocks, degrading the safety margin to $z/\xi \approx 2.9$.

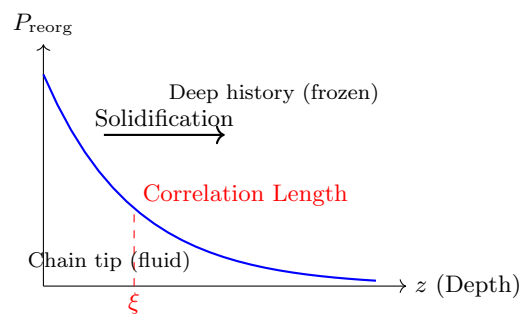

Figure 5: **Correlation decay.** The probability of reorganization decays exponentially with depth z. The system transitions from a "fluid" phase at the tip to a "frozen" solid phase in the bulk. The characteristic length scale ξ defines probabilistic finality.

C. Nucleation of a False History

An attack on the chain is analogous to the **nucleation of a false vacuum bubble**. An attacker attempting to rewrite z blocks must create a "domain" of length z with a divergent spin configuration (an alternative history).

In standard statistical mechanics, the probability of spontaneous nucleation is exponentially suppressed by the free-energy cost $\Delta\mathcal{F}$ of the domain: $P_{\text{nucleation}} \propto e^{-\Delta\mathcal{F}/k_B T_{\text{eff}}}$. By equating this with our macroscopic reorganization probability derived above ($P_{\text{reorg}} = e^{-z/\xi}$), we can extract the exact thermodynamic potential barrier $\Delta\mathcal{F}(z)$ required to rewrite history:

$$\Delta\mathcal{F}(z) = z\left(\frac{k_B T_{\text{eff}}}{\xi}\right) = z \cdot k_B T_{\text{eff}} \ln\left(\frac{p}{q}\right). \tag{49}$$

This formulation shows that the free-energy barrier to rewrite history grows approximately linearly with depth z, while being dynamically scaled by the effective temperature of the network and the mass gap. Immutability is therefore not absolute, but **thermodynamically robust**. Rewriting history is not impossible; rather, in this effective description it becomes exponentially disfavored by macroscopic dissipation, as the external work required to overcome $\Delta\mathcal{F}(z)$ quickly diverges beyond physically realizable bounds.

D. Forks as Absorbing-State Transitions (Directed Percolation)

To formalize the suppression of alternative histories, we map the blockchain growth process to a 1+1D Directed Percolation (DP) model. We define the spatial axis as the topology of network nodes and the directed temporal axis as the block height z. In this framework, a competing fork represents an "active" state propagating through the network, while the canonical longest chain acts as the **absorbing state**.

Since nodes default to the heaviest chain, abandoned forks cannot spontaneously reactivate; they are permanently absorbed into the topological vacuum. We define the control parameter for this transition as the hashrate differential $\Delta p = p - q$. Since $p + q = 1$, this differential is strictly proportional to the standard proximity parameter $|p - p_c|$ (with critical threshold $p_c = 0.5$) used to define absorbing-state transitions in Directed Percolation universality classes.

If the extinction of competing forks satisfies the conditions for the directed-percolation universality class—short-range interactions, a unique absorbing state, and no additional conservation laws [23]—then the survival probability $\mathcal{P}(z)$ of a false history to depth z is expected to follow the scaling form

$$\mathcal{P}(z) \propto z^{-\delta} \exp\left(-\frac{z}{\xi_\parallel}\right), \tag{50}$$

where $\xi_\parallel$ is the parallel correlation length, diverging near the critical point as $\xi_\parallel \propto |\Delta p|^{-\nu_\parallel}$.

When the honest network commands a majority ($p > q$), the system operates effectively in the **subcritical regime** of directed percolation. The active fork density exponentially decays toward zero. The "6-Block Rule" is thus physically equivalent to the assertion that at depth $z \approx 6$, the survival probability of the active state has been overwhelmingly attenuated by the absorbing boundary of the canonical history. If an attacker achieves $q \geq p$, the system crosses the percolation threshold into the supercritical active phase; false histories percolate infinitely through z, topological finality collapses, and the Markov blanket is irrevocably breached.

We note two caveats regarding the DP mapping. First, the Difficulty Adjustment Algorithm introduces a long-range temporal feedback that is absent in standard short-range DP models; this may alter the critical exponents near the transition. Second, the absorbing state (the canonical chain) is itself dynamically extending, unlike a fixed absorbing boundary. These deviations suggest that directed percolation is best viewed here as an organizing analogy, and possibly as a candidate modified universality class only after dedicated finite-size scaling and microscopic validation. We therefore leave precise universality-class identification for future investigation.

V. INFORMATION ENTROPY AND MARKOV BLANKET

The physical stability of the ledger depends on how the protocol processes and stores information against the effective limits of the communication channel, parameterized here by C_{eff}. Given the thermal percolation constraint defined previously ($V_B \ll V_{\text{B,crit}}$), the network cannot continuously propagate and verify the entirety of the historical "volume." In our effective description, the protocol mitigates this bottleneck by projecting history onto a reduced state description, which we interpret heuristically as an effective **Markov blanket**. The term is used here in a coarse-grained informational sense, not in the strict Bayesian-network sense of exact conditional-independence factorization.

A. The Volume, the Markov Blanket, and RG Flow

A Markov blanket [24] screens internal from external degrees of freedom. In Nakamoto consensus the UTXO set plays this role: the *volume* $\mathcal{V}$ is the full blockchain (all past transactions plus the cumulative PoW action $\mathcal{S}_{\text{PoW}}$), and the *blanket* $\partial\mathcal{V}$ is the UTXO set—a compressed active-state description sufficient for validation without history replay. Finite bandwidth prevents real-time processing of the full history. Block validation therefore acts as a non-invertible decimation operator that we interpret, at level (iii), as analogous to a renormalization-group coarse-graining step $\mathcal{R}$:

$$\partial\mathcal{V}_t = \mathcal{R}(\mathcal{V}_{t-1} \cup \mathcal{B}_t), \tag{51}$$

where $\mathcal{B}_t$ is the transaction set in the new block. Because this many-to-one map is irreversible, $S(\mathcal{V}) \gg S(\partial\mathcal{V})$, constituting a dissipative compression of the validation state. Separately, the protocol's block-size cap constrains the per-block exchange volume V_B to remain below the critical threshold $V_{\text{B,crit}}$ [Eq. (27)]; the RG-like compression ensures that the *state* needed to enforce this constraint remains tractable.

B. Shannon vs. Thermodynamic Entropy

The Markov-blanket projection resolves the apparent tension between Shannon [7] and Landauer [4] entropy. At the hardware level the dissipated heat flux is bounded by

$$\dot{Q} \;\geq\; k_B T_{\text{sub}} \ln 2 \, \frac{dS_{\text{Shannon}}}{dt}, \tag{52}$$

where S_{Shannon} is measured in bits and T_{sub} is the physical substrate temperature (distinct from the macroeconomic bath temperature T_{bath} of Sec. VII D).

Second, to rigorously ground this constraint at the *macro-canonical consensus level*, we invoke a generalized

form of the fluctuation theorem (derived from Jarzynski's equality [25] for information processing). The average irreversible macroscopic work $\langle W_{\rm diss}\rangle$ provided by the network to synchronize a new block across the topological graph must satisfy

$$\langle W_{\rm diss}\rangle \geq k_B T_{\rm eff} \ln(2)\, \Delta S_{\rm Shannon}\,, \tag{53}$$

where $\Delta S_{\rm Shannon}$ represents the spatial Shannon uncertainty (in bits) injected by the flow of unconfirmed new transactions. For a node to validate the network state autonomously, this massive macroscopic work—governed by the effective information temperature $T_{\rm eff}$—dissipates entropy so that uncertainty regarding the UTXO boundary state is effectively brought to zero ($S_{\rm Shannon} = 0$).

C. The UTXO Set as a Physical Maxwell's Demon

The Mass-Energy-Information equivalence [26] becomes manifest when modeling the network as a macroscopic Maxwell's Demon. As Bennett demonstrated [3], a Demon must eventually erase its memory, paying a thermodynamic cost governed by Landauer's Principle [4]:

$$W_{\rm erase} \geq k_B T_{\rm sub} \ln(2)\,, \tag{54}$$

per bit of erased information. The network acts as a decentralized Demon, sorting the high-entropy Mempool "gas" across the barrier. Crucially, the thermodynamic cost is not merely storing the accepted history $\mathbf{B}_A$, but the relentless evaluation and active rejection of competing histories. To reset the UTXO boundary for the next epoch, the network must continuously flush these discarded states. Rather than operating near the minimal energetic bounds suggested by Landauer's principle, the protocol utilizes the DAA to artificially inflate this required dissipation by dozens of orders of magnitude. To reduce uncertainty and synchronize a canonical block across the network, the protocol dissipates energy at a rate

$$\dot{Q}_{\rm diss} \sim \frac{\bar{E}_{\rm eff}}{\tau_B}\,. \tag{55}$$

In practice, this dissipation is vastly larger than the minimal Landauer scale,

$$\dot{Q}_{\rm diss} \gg k_B T_{\rm sub} \ln(2)\, \frac{dS_{\rm Shannon}}{dt}\,. \tag{56}$$

This massive divergence allows us to formally define the **information-thermodynamic efficiency** $\eta_{\rm info}$ of the Nakamoto Demon:

$$\eta_{\rm info} \equiv \frac{\dot{W}_{\rm Landauer}}{\dot{Q}_{\rm diss}} \approx \frac{k_B T_{\rm sub} \ln(2) \frac{dS_{\rm Shannon}}{dt}}{\bar{E}_{\rm eff}/\tau_B} \to 0\,. \tag{57}$$

Because the DAA scales the macroscopic energy requirement $\bar{E}_{\rm eff}$ with global exergy availability, while the information processing rate $\frac{dS_{\rm Shannon}}{dt}$ remains strictly bounded by $C_{\rm eff}$, the protocol is driven toward an asymptotic efficiency of zero.

This formulation motivates the phenomenological inertia scale introduced in Sec. VIII B: **macroscopic thermodynamic cost acts as an effective security mechanism**. Within the coarse-grained description, immutability is sustained because reversing the ledger would require re-supplying at least the cumulative proof-of-work exergy already dissipated, a cost that grows linearly with chain depth. At the phenomenological level this makes deep reorganizations exponentially suppressed, in direct analogy with nucleation barriers in first-order phase transitions.

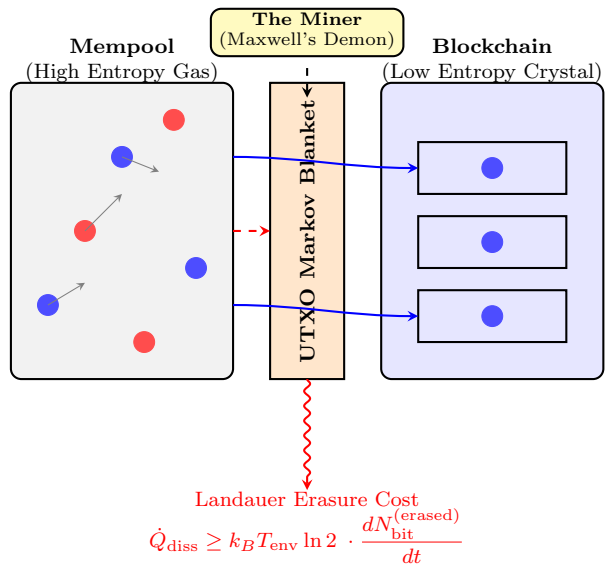

Figure 6: **Nakamoto consensus as an effective Maxwell-demon-like information engine.** The network operates in a Maxwell-demon-like manner only in an *effective coarse-grained* sense: candidate transactions are filtered through the UTXO boundary, valid state updates are retained, and rejected or overwritten alternatives (including orphaned branches and invalid transactions) contribute to logical irreversibility. The displayed Landauer expression should therefore be read as an analogy-level lower bound on information-processing dissipation rather than as a claim of literal device-level reversibility accounting. Here $T_{\rm env}$ denotes the environmental (substrate) temperature of the mining hardware.

D. Blanket Saturation and Channel Limit

The security of this effective Markov blanket is linked, within the present coarse-grained description, to the network's finite information-transport capacity. State updates $\partial\mathcal{V}$ (new blocks of volume V_B) must pass through the effective communication channel $C_{\rm eff}$. As the boundary update burden approaches the transport threshold ($V_B \to V_{\rm B,crit}$), the assumption of globally synchronized state estimation degrades: nodes can no longer reconcile the UTXO surface within the allotted time τ_B. In that regime, local Shannon uncertainty increases, the effective temperature $T_{\rm eff}$ rises toward its critical value, and consensus may fragment into competing domains. History

rigidity is therefore maintained only while the boundary update size remains below the network's coarse-grained dissipation capacity.

VI. TOPOLOGICAL STABILITY OF THE NETWORK GRAPH

While the 1D Directed Percolation model isolates longitudinal stability (temporal depth), robustness against partition attacks requires analyzing transverse stability—the long-range phase order across the spatial peer-to-peer network. We model this using the **XY Model** on a **"small-world"** graph [27].

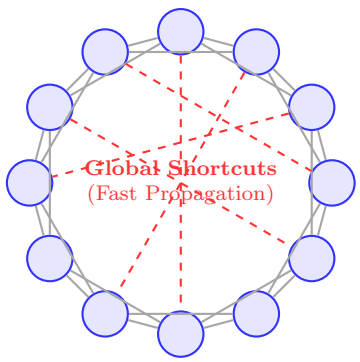

Figure 7: The "Small-World" topology of a distributed consensus network. Local links (gray) ensure redundancy and fault tolerance, while long-distance connections (dashed red) allow information to traverse the network in few hops, minimizing propagation delay.

This architecture is characterized by two fundamental physical and mathematical properties:

1. **A high clustering coefficient:** Nodes form highly interconnected local clusters, ensuring massive information redundancy at the regional level.
2. **A very short average path length (shortcuts):** A few long-distance random connections link distant clusters. The average shortest path scales as $D_{\text{hop}} \sim \ln N_{\text{nodes}}$, where N_{nodes} is the number of reachable peers. It is this logarithmic diameter that enters the latency model $\tau_L = D_{\text{hop}}\left[\tau_{\text{link}} + V_B\left(C_{\text{node}}^{-1} + \gamma\right)\right]$, where $\gamma\,[\text{s byte}^{-1}]$ is a per-byte validation overhead and the bracketed quantity is the per-hop delay (propagation plus bandwidth plus validation), and, through Eq. (27), sets the critical block volume.

A. The Consensus Phase Field

We assign a continuous phase variable $\theta_i \in [0, 2\pi)$ to each node i, representing its local view of the ledger tip. Interaction energy between peers seeks to minimize phase differences (consensus):

$$\mathcal{H}_{XY} = -\bar{J} \sum_{\langle i,j\rangle \in \mathcal{M}} \cos(\theta_i - \theta_j)\,, \tag{58}$$

where $\mathcal{M}$ represents the specific subgraph of active mining nodes (we denote the Hamiltonian as $\mathcal{H}_{XY}$ to explicitly distinguish it from the macroscopic hashrate $H(t)$). Crucially, non-mining nodes (which merely relay data) do not contribute to the topological rigidity of the consensus; they act as passive observers adopting the local phase field.

To ensure internal consistency of the model, we define the mean coupling constant $\bar{J}$ not as an arbitrary free parameter, but as the average local action density injected into the active network:

$$\bar{J} \approx h_N(t) \cdot \langle H_i \rangle_{\mathcal{M}}\,. \tag{59}$$

$\langle H_i \rangle_{\mathcal{M}}$ represents the mean absolute hashrate (in s^{-1}) of an active mining cluster. This formulation explicitly bridges the micro-canonical and macro-canonical scales: the microscopic switching time of a silicon logic gate (τ_{hash} embedded within h_N) serves as the fundamental quantum of action that physically finances the macroscopic phase alignment (θ) of decentralized nodes across the globe. It directly links the topological rigidity of consensus (vortex resistance) to actual hardware-level energy dissipation.

- **Ground State ($\nabla\theta = 0$):** Global Consensus. All nodes agree on the tip.
- **Excited State ($\nabla\theta \neq 0$):** A Fork. The network is split into domains with different phases.

B. Forks as Topological Defects (Vortices)

In the two-dimensional P2P overlay, thermal fluctuations generate topological defects (**vortices**)—closed loops of nodes maintaining divergent chain-tip views, analogous to the localized phase alignment in the Axelrod model [28] and the synchronization boundaries in epidemic-consensus models [29]:

$$\oint_{\mathcal{C}} \nabla\theta \cdot d\mathbf{l} = 2\pi n, \quad n \in \mathbb{Z}\,, \tag{60}$$

where the continuum integral is understood as the mean-field limit of the discrete graph sum over directed edges $\langle i,j\rangle$ forming a closed loop $\mathcal{C}$:

$$\sum_{\langle i,j\rangle \in \mathcal{C}} \mathcal{W}(\theta_j - \theta_i) = 2\pi n\,, \tag{61}$$

where $\mathcal{W}(x) \equiv x - 2\pi\lfloor (x+\pi)/(2\pi)\rfloor$ maps phase differences to the principal branch $[-\pi, \pi)$. This suggests a discrete-XY-like effective description on the communication graph. Here, θ_i represents a coarse-grained local chain-tip phase at node i. The $U(1)$-like symmetry invoked in the BKT analogy should be understood as emergent and coarse-grained, not as a literal microscopic rotor variable: local chain-tip views can deviate

continuously in the effective phase description, while the dominant penalty depends on phase gradients across the network. If $n \neq 0$, the loop encloses a singularity (a persistent fork). Because the energetic cost of such a defect depends on the effective dimensionality of the network; isolated forks are suppressed differently depending on the structural health of the P2P graph. We emphasize that the BKT analogy requires a continuous $U(1)$-like phase symmetry. In the pure longest-chain rule, tip selection is binary ($\mathbb{Z}_2$); however, during active propagation delays each node can be represented as carrying an effective fractional commitment across competing tips, weighted by first-seen arrival times. This motivates an approximate continuous phase description $\theta_i \in [0, 2\pi)$ at the coarse-grained level. The resulting continuous limit, relevant in the mempool regime, is a level-(iii) interpretive assumption (see Sec. I), not a literal microscopic statement about the protocol state space.

C. Effective Dimensionality and the Kosterlitz–Thouless Bound

As a *heuristic, idealized* description of spatial consensus loss under severe network partitioning, we analyze vortex dynamics using the **Kosterlitz–Thouless (KT)** formalism applied to the limiting case of a 2D-reduced propagation layer. The quantitative accuracy of KT scaling on a partitioned scale-free graph remains an open question; we present this analysis as a bounding estimate rather than an exact prediction. Before applying 2D topological physics to a scale-free graph, it is necessary to resolve an apparent topological paradox. The Mermin–Wagner Theorem [19] states that spontaneous breaking of a continuous symmetry is impossible at finite temperature for *equilibrium* systems of dimension $d \leq 2$. While the network operates as a Non-Equilibrium Steady State (NESS) driven by active noise—a condition known to allow behavior that can circumvent standard Mermin–Wagner expectations in certain active-matter systems—the primary topological defense mechanism lies in the dynamic effective dimensionality d_{eff} of the graph, extracted from the scaling law linking the number of accessible nodes $\mathcal{N}_{\text{nodes}}(r)$ within a topological radius r (number of hops):

$$\mathcal{N}_{\text{nodes}}(r) \propto r^{d_{\text{eff}}} . \tag{62}$$

In a healthy state, the P2P graph exhibits strong small-world connectivity, which we represent phenomenologically by a large effective spectral dimensionality ($d_{\text{eff}} \gg 1$), defined via the return-probability exponent of a random walk on the P2P graph [30], owing to global intercontinental shortcuts. However, when a severe network partition occurs (e.g., the isolation of a continental grid), the routing topology severely degrades. Stripped of its global shortcuts, the propagation layer is forced to embed much more closely onto the physical 2D surface of the Earth. Under massive link failure, the effective propagation topology loses its small-world shortcuts. As an idealized limiting case we model the residual graph as a two-dimensional lattice with spatial dimension $d_s \approx 2$, consistent with the geographic embedding of surviving continental links [31, 32]. Accordingly, a severely fragmented network may be approximated, as an idealized limiting case, by an effectively two-dimensional propagation layer.

If severe degradation of the communication graph admits a coarse-grained effective two-dimensional phase description, then the energy of an isolated vortex-like defect (here interpreted as a partitioned fork) grows logarithmically with network size L:

$$E_{\text{vortex}} \approx \pi \bar{J} \ln\left(\frac{L}{a_0}\right) , \tag{63}$$

where a_0 is the lattice constant (the minimum discrete distance, typically one hop), and $\bar{J}$ effectively absorbs the macroscopic spin-wave stiffness of the network graph.

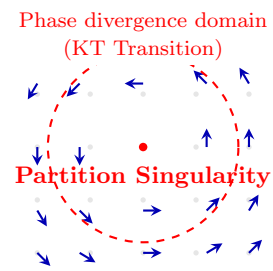

Figure 8: **Topological defect in the P2P network.** Arrows: node consensus phases θ_i, defined as the continuous weight each node assigns to the leading chain tip. A free vortex ($n=1$) destroys long-range order, preventing global consensus. We note that the continuous U(1) phase interpretation requires nodes to maintain fractional commitment across competing tips; the pure binary (longest-chain) rule would instead yield an Ising ($\mathbb{Z}_2$) symmetry for which BKT physics does not apply. The continuous limit is justified in the mempool regime where nodes simultaneously relay and validate multiple candidate blocks.

Simultaneously, the vortex core can be located on any of the $(L/a_0)^2$ sites of the two-dimensional lattice. The system thus acquires a configurational entropy:

$$S_{\text{vortex}} \approx k_B \ln\left[\left(\frac{L}{a_0}\right)^2\right] = 2k_B \ln\left(\frac{L}{a_0}\right) . \tag{64}$$

Consensus stability against free vortex formation is governed by the variation in free energy $\Delta\mathcal{F} = E_{\text{vortex}} - T_{\text{eff}} S_{\text{vortex}}$:

$$\Delta\mathcal{F} = 2k_B (T_{\text{KT}} - T_{\text{eff}}) \ln\left(\frac{L}{a_0}\right) . \tag{65}$$

This balance equation identifies an effective critical partition temperature $T_{\text{KT}} = \pi\bar{J}/(2k_B)$ within the present KT-inspired phenomenology, representing the topological unbinding threshold at which global phase coherence is expected to fail. In this effective description, survival of the canonical ledger truth is therefore modeled as a topological phase-transition problem:

- **Ordered ($T_{\rm eff} < T_{\rm KT}$):** $\Delta\mathcal{F} > 0$; free-vortex creation costs infinite energy as $L \to \infty$. Vortices bind in neutral pairs; global consensus holds.
- **Partitioned ($T_{\rm eff} > T_{\rm KT}$):** $\Delta\mathcal{F} < 0$; isolated vortices proliferate, destroying phase coherence and fracturing the ledger (Fig. 8).

The small-world topology confines the network to $T_{\rm eff} \ll T_{\rm KT}$, preventing vortex unbinding.

VII. NON-EQUILIBRIUM DYNAMICS: EXERGETIC EXPANSION AND THE "HALVING" QUENCH

While Sec. VI addresses spatial stability, the network must also survive severe temporal shocks. As an open, non-equilibrium system coupled to the global energy infrastructure, the Nakamoto protocol continuously scavenges **exergy** (e.g., stranded energy or gas flaring) to sustain macroscopic order.

Consequently, the "Halving" is not a simple parametric update, but a *thermodynamic quench* superimposed on an expanding, non-stationary potential.

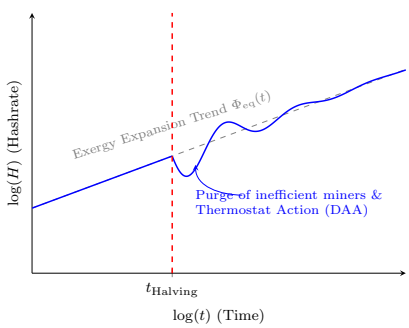

Figure 9: **Quench on non-stationary background.** Order parameter Φ is sustained by secular growth (exergy hunting). The Halving imposes an abrupt perturbation. Miner inertia and thermostat intervention (DAA) create a transient plateau or dip (damped relaxation) before the secular growth trend resumes.

A. The Exergy-Driven Potential

The homogeneous effective potential $\mathcal{U}(\Phi)$ is driven by the total **exergetic chemical potential** $\mu(t)$, defined as the macroscopic pool of available thermodynamic work (e.g., stranded energy in Joules). This potential features a secular growth component decomposed as

$$\mu(t) = \mu_{\rm subsidy}(t) + \mu_{\rm exergy}(t)\,. \tag{66}$$

The subsidy acts as a cascaded thermodynamic quench, formalized via a discrete floor function: $\mu_{\rm subsidy}(t) = \mu_{\rm genesis} \cdot 2^{-\lfloor t/\tau_H \rfloor}$. For a single Halving at t_H, this reduces to a Heaviside step function where the economic reward acts as a proxy for accessible Joules.

To respect planetary boundaries, the exergy reservoir $\mu_{\rm exergy}(t)$ captured by the network is ultimately bounded by the biosphere's maximum accessible exergy $\mu_{\rm max}$. Defining the effective net exergy injection rate as $P_{\rm in}(t) \equiv \dot{\mu}_{\rm exergy}(t)$.

a. Notation: local vs. global order parameter. We use lower-case $\phi(x,t)$ for the spatially-resolved, dimensionless Ginzburg–Landau field and upper-case $\Phi(t) \equiv \mathcal{E}_{\rm exergy}(t)/\mathcal{E}_0$ for the dimensionless global-mode order parameter, where $\mathcal{E}_0$ [J] is a reference exergy scale (e.g. the block-reward exergy at a chosen epoch). All stochastic-dynamics equations in Secs. VII Dff. are formulated in terms of the dimensionless Φ.

At the coarse-grained level, the exergy reservoir dynamics are assumed to obey the logistic first-order ODE

$$\dot{\mu}_{\rm exergy}(t) = r_{\rm tech}(t)\,\mu_{\rm exergy}(t)\left(1 - \frac{\mu_{\rm exergy}(t)}{\mu_{\rm max}}\right). \tag{67}$$

Here $\mu_{\rm exergy}(t)$ [J] denotes the coarse-grained exergy stock effectively committed to the network, so that $\dot{\mu}_{\rm exergy}$ [W] is the corresponding instantaneous exergy injection power. Dimensional check: the logistic factor $(1 - \mu_{\rm exergy}/\mu_{\rm max})$ is dimensionless, hence $[\dot{\mu}_{\rm exergy}] = [r_{\rm tech}]\,[\mu_{\rm exergy}] = {\rm s}^{-1} \cdot {\rm J} = {\rm W}$. The net exergy injection power into the consensus layer is therefore $P_{\rm in}(t) \equiv \dot{\mu}_{\rm exergy}(t)$, which equals the gross exogenous supply modulated by the logistic saturation factor $(1-\mu_{\rm exergy}/\mu_{\rm max})$. Thus $P_{\rm in}$ is not a free external parameter but a self-consistent consequence of Eq. (67). The growth rate $r_{\rm tech}(t)$ [$\rm s^{-1}$] is the effective technology-driven rate, and $\mu_{\rm max}$ [J] is the planetary exergy ceiling. Empirically, $r_{\rm tech}(t)$ exhibits diminishing returns (approximately $\propto \nu_H/t$ over the fitted secular regime), which suppresses indefinite exponential acceleration of $\mu_{\rm exergy}(t)$ as the system matures.

Excluding the pre-2010 transient phase, the network operates far from planetary saturation ($\mu_{\rm exergy} \ll \mu_{\rm max}$). The time-dependent innovation rate forces an effective temporal power-law attractor (Fig. 10):

$$H(t) = A \cdot \left(\frac{t}{t_0}\right)^{\nu_H}, \tag{68}$$

where t is the time since genesis and $t_0 = 1\,$day is the characteristic reference time. Regression in log–log space over the fitted post-transient window yields an effective scaling exponent $\nu_H = 11.59 \pm 0.31$ and an amplitude $A = (6.7 \pm 3.1) \times 10^{-35}$ TH/s (95% confidence intervals from ordinary least squares on the log-transformed data). We emphasize that ν_H is a finite-window empirical scaling exponent, not a claim of asymptotic critical scaling over the full historical evolution.

We conduct a preliminary model comparison on $N = 5{,}114$ daily post-2010 log-hashrate observations using Bayesian Information Criterion (BIC). Candidate models are: (i) pure exponential, with 2 free parameters; (ii) power-law, with 2 free parameters; (iii) stretched exponential ($\mu(t) = \mu_0 \exp\left[(\tau/\tau_c)^\beta\right]$), with 3 free parameters. BIC scores are computed in log-space using the

Gaussian likelihood. Results: $\Delta\mathrm{BIC}_{\mathrm{exp-PL}} \approx +87$ (exponential disfavored) and $\Delta\mathrm{BIC}_{\mathrm{SE-PL}} \approx +5$ (stretched exponential moderately disfavored relative to power law), both relative to the power-law model [33]. We stress that this comparison selects the most parsimonious functional form within the observed 2010–2026 window; it does not establish an asymptotic law, universality class, or critical exponent. We caution that the comparison is performed in log-space; the large exponent ν_H means the power-law form is effectively a reparametrization of super-exponential growth over the observed range. The detailed fit comparison, including residual diagnostics, is available in the accompanying repository [34]. These results support the power-law form as a parsimonious descriptive fit over the observed interval in log-space, but do not establish an exact asymptotic law and do not exclude more complex sub-decadal dynamics. The large fitted exponent $\nu_H \approx 11.59$ should be interpreted as an effective finite-window scaling exponent; it does not correspond to any known equilibrium or non-equilibrium universality class and should not be read as a critical exponent in the renormalization-group sense. Its magnitude reflects the sustained super-exponential hardware investment over 2010–2024 and may decrease substantially as the network approaches resource saturation.

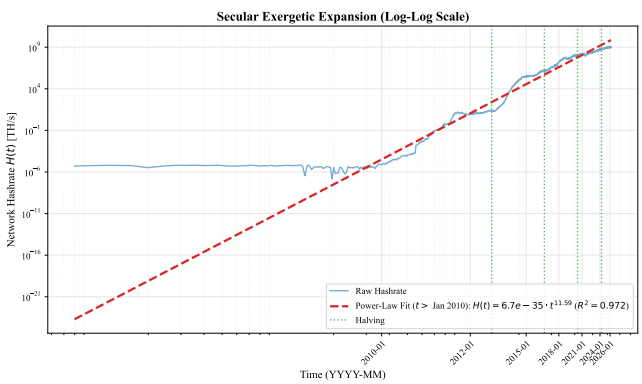

Figure 10: **Secular exergetic expansion.** The network's hashrate $H(t)$ aligns with a power-law attractor $H(t) \propto (t - t_0)^{\nu_H}$ with $\nu_H \approx 11.59$ ($R^2 = 0.972$), where t_0 is a fitted time origin. This expansion is consistent with the qualitative expectations of the Maximum Power Principle, insofar as the network continues to expand exergy capture despite periodic Halving quenches.

B. Non-Equilibrium Stochastic Dynamics and FDT Violation

Assuming the network operates deep within the ordered phase ($T_{\mathrm{eff}} \ll T_{\mathrm{KT}}$), the phase-like mode is taken to remain effectively pinned ($\nabla\theta \approx 0$), allowing us to reduce the spatial field to a real global temporal order parameter,

$$\Phi(t) \equiv \langle \phi(x,t) \rangle_x \,. \tag{69}$$

By treating $\Phi(t)$ as the normalized deviation (H/H_{eq}) around the moving adiabatic equilibrium, the spatially integrated homogeneous potential $\mathcal{U}(\Phi)$ becomes quasi-stationary during the short temporal window of a quench. Trapped in the positive vacuum ($+\phi_0$), $\Phi(t)$ is effectively confined to $\mathbb{R}^+$. Because the system operates deep within this well ($\Phi \approx \Phi_0 \gg 0$), the probability of zero-crossing is exponentially suppressed. This justifies using standard additive noise with a formal reflective boundary at $\Phi = 0$ to prevent unphysical negative hashrate excursions.

To capture the network's pre-heating phase driven by forward-looking capital expenditure, we inject a phenomenological anticipatory driving force $\mathcal{A}(t)$ into the Langevin equation for the global mode. This term encodes observed miner behavior—hardware pre-deployment in anticipation of subsidy changes—and is not derived from the thermodynamic framework itself:

$$\frac{d\Phi(t)}{dt} = -\Lambda_0 \frac{\partial \mathcal{U}(\Phi)}{\partial \Phi} + \mathcal{A}(t) + \zeta(t)\,, \tag{70}$$

where

$$\mathcal{U}(\Phi) = \alpha_{\mathrm{h}}(T_{\mathrm{eff}})\,\Phi^2 + b_{\mathrm{h}}\,\Phi^4 + \mathcal{U}_{\mathrm{ext}}(t), \tag{71}$$

where $\mathcal{U}_{\mathrm{ext}}(t)$ accounts for secular exergy expansion. This identifies the effective "consensus stiffness" $\kappa_\Phi = \partial^2\mathcal{U}/\partial\Phi^2$ as the fundamental restorative scale preventing hashrate evaporation.

The relaxation mobility Λ_0 $[\mathrm{J}^{-1}\,\mathrm{s}^{-1}]$ sets the intrinsic timescale on which the order parameter responds to potential gradients. Its inverse, Λ_0^{-1}, carries dimensions of $\mathrm{J}\cdot\mathrm{s}$ and represents the effective inertia of the global mode against restoring forces. An order-of-magnitude estimate follows from requiring the deterministic relaxation time $\tau_{\mathrm{relax}} \sim (\Lambda_0\,|\partial^2\mathcal{U}/\partial\Phi^2|_{\Phi_0})^{-1}$ to be comparable to the empirically observed DAA adjustment timescale ($\sim 2\,\mathrm{weeks} \approx 1.2\times10^6\,\mathrm{s}$). With $|\partial^2\mathcal{U}/\partial\Phi^2|_{\Phi_0} = 2\kappa_{\mathrm{marginal}} \sim \mathcal{O}(10^{15})\,\mathrm{J}$ (from the network's total per-epoch dissipation scale), this yields $\Lambda_0 \sim \mathcal{O}(10^{-21})\,\mathrm{J}^{-1}\,\mathrm{s}^{-1}$. Calibration against post-Halving relaxation curves would refine this estimate. As a self-consistency check, the diffusion coefficient $D = \Lambda_0 k_B T_{\mathrm{act}}$ implied by this estimate should produce variance growth $\langle(\delta\Phi)^2\rangle \sim 2D\,\tau_{\mathrm{relax}}$ comparable to the empirical post-Halving variance spikes ($\sim$ 10–50%), which is satisfied for $k_B T_{\mathrm{act}} \sim \mathcal{O}(10^{15})\,\mathrm{J}$.

Here α_{h} and b_{h} carry dimensions of absolute energy [J], representing the spatial integration of the local free-energy density coefficients a and b $[\mathrm{J\,m}^{-1}]$ over the effective network diameter L_{net} [m]:

$$\alpha_{\mathrm{h}} = a \cdot L_{\mathrm{net}}\,, \qquad b_{\mathrm{h}} = b \cdot L_{\mathrm{net}}\,. \tag{72}$$

This bridges the continuous 1D field theory of Sec. III B to the lumped-parameter global dynamics.

The anticipatory drift $\mathcal{A}(t)\,[\mathrm{s}^{-1}]$ is defined as the mobility-weighted, exponentially discounted expectation of the future rate of change of the subsidy exergy over a

finite planning horizon $\tau_{\rm plan}$ [s]:

$$\mathcal{A}(t) = -\Lambda_0 g_{\rm ant} \int_t^{t+\tau_{\rm plan}} e^{-\lambda(t'-t)} \mathbb{E}\left[\frac{\partial \mu_{\rm subsidy}(t')}{\partial t'} \middle| \mathcal{F}_t \right] dt' \,. \tag{73}$$

Here $g_{\rm ant}$ [1] is a dimensionless anticipation coupling constant, $\mathcal{F}_t$ denotes the natural filtration of market information up to time t, $\lambda\,[{\rm s}^{-1}]$ is the exponential discount rate of future exergy flows, and $\Lambda_0\,[{\rm J}^{-1}{\rm s}^{-1}]$ is the kinetic mobility introduced in the Langevin sector. The integrand has dimensions $[{\rm J\,s}^{-1}]$, so after time integration one obtains an energy scale [J], and multiplication by Λ_0 yields $[{\rm s}^{-1}]$, as required for consistency with the drift terms in the Langevin equation governing the dimensionless order parameter $\Phi(t)$. Because the Halving subsidy collapse is a deterministic downward step function, its temporal derivative constitutes a negative Dirac delta $-\Delta\mu\,\delta(t'-t_H)$, where $\Delta\mu\,[{\rm J}]$ is the magnitude of the subsidy drop. Because $\Delta\mu > 0$ is defined as the magnitude of the anticipated subsidy reduction, the driving term enters with the sign required to bias the global mode upward before the event. Integrating over the delta function then yields a positive macroscopic anticipatory pulse leading up to t_H, provided the Halving event falls within the planning horizon ($t_H - \tau_{\rm plan} \le t \le t_H$):

$$\mathcal{A}(t) \approx \Lambda_0\, g_{\rm ant}\, \Delta\mu\, e^{-\lambda(t_H-t)}\, \Theta(t_H - t)\, \Theta(t - t_H + \tau_{\rm plan})\,, \tag{74}$$

where the Heaviside factors restrict the pulse to the anticipatory window $t_H - \tau_{\rm plan} \le t \le t_H$, $g_{\rm ant}$ is a dimensionless anticipation coupling, λ $[{\rm s}^{-1}]$ sets the anticipation time-scale, and the effective planning horizon is $\tau_{\rm plan} \simeq \lambda^{-1}$. Dimensional consistency: since the global mode Φ is dimensionless and the homogeneous potential $\mathcal{U}(\Phi)$ carries dimensions of energy [J], the kinetic mobility requires $[\Lambda_0] = {\rm J}^{-1}{\rm s}^{-1}$ (cf. Appendix VIII C). The anticipation coupling $g_{\rm ant}$ must therefore be strictly dimensionless ([1]), so that the drift magnitude $[\mathcal{A}] = [\Lambda_0]\,[g_{\rm ant}]\,[\Delta\mu] = {\rm J}^{-1}{\rm s}^{-1}\cdot 1\cdot {\rm J} = {\rm s}^{-1}$, perfectly matching the required dimensions for the temporal evolution of the dimensionless order parameter Φ in Eq. (70).

Physically, $\mathcal{A}(t)$ represents a mobility-weighted exergy run-up that biases the global mode toward faster block-production conditions as miners reposition capital ahead of the subsidy quench.

Operating far from equilibrium, the network sustains a strictly positive irreversible entropy production rate. Within the present information-thermodynamic coarse graining, each erased bit of orphaned-chain state is modeled as dissipating at least $k_B T_{\rm sub}\ln 2$ of heat under the Landauer–Bennett bound. Dividing the corresponding minimal heat rate by $T_{\rm sub}$ yields the entropy-production bound

$$\dot{S}_{\rm irrev} \;\ge\; \frac{k_B T_{\rm sub}\ln 2}{T_{\rm sub}}\,\frac{dN_{\rm bit}^{\rm (erased)}}{dt} \;=\; k_B \ln 2\,\frac{dN_{\rm bit}^{\rm (erased)}}{dt} \;>\; 0\,, \tag{75}$$

where $N_{\rm bit}^{\rm (erased)}$ counts discarded orphan-block bits (dimensionless, in natural units where 1 bit = 1), and the first equality follows from the Landauer bound $\delta Q \ge k_B T_{\rm sub}\ln 2$ per erased bit divided by the substrate temperature $T_{\rm sub}$.

This persistent dissipation ($\dot{S}_{\rm irrev} > 0$) signals violation of the equilibrium Fluctuation–Dissipation Theorem (FDT). To formalize this Non-Equilibrium Steady State (NESS), we quantify the deviation using the correlation function $C_\Phi(t,t') = \langle \delta\Phi(t)\delta\Phi(t')\rangle$ and the linear response function $R(t,t')$ to a localized exergy injection $h(t')$. This yields the Fluctuation–Dissipation Ratio (FDR) $X(t,t')$, with the linear response kernel defined by $\delta\langle\Phi(t)\rangle = \int dt'\, R(t,t')\, h(t')$, where the conjugate source field $h(t')$ is defined as a local modulation of the exergy injection field, carrying units of energy ($[h] = {\rm J}$). From the convolution $\delta\langle\Phi(t)\rangle = \int dt'\, R(t,t')\, h(t')$, and since $[\Phi] = 1$, we obtain $[R] = [\Phi]/([{\rm s}]\cdot[h]) = {\rm J}^{-1}{\rm s}^{-1}$. Using the present sign and time-derivative convention, we write

$$R(t,t') = \frac{X(t,t')}{k_B T_{\rm bath}}\,\frac{\partial C_\Phi(t,t')}{\partial t'}\,\Theta(t-t')\,. \tag{76}$$

Here $X = 1$ recovers the equilibrium FDT under the same convention. Here

$$C_\Phi(t,t') \equiv \langle \delta\Phi(t)\,\delta\Phi(t')\rangle \tag{77}$$

is the two-time connected correlation function of the coarse-grained order parameter, and $X(t,t')$ is the fluctuation–dissipation ratio (FDR), defined as $X(t,t') \equiv T_{\rm eff}(t,t')/T_{\rm act}$, where $T_{\rm eff}$ is the effective temperature extracted from the ratio of the response function to the time derivative of the correlation function. In thermal equilibrium $X = 1$; persistent deviation $X \ne 1$ signals non-equilibrium driving, as expected for the consensus layer under continuous exergy injection.

The response kernel $R(t,t') \equiv \delta\langle\Phi(t)\rangle/\delta h(t')$ carries units $[{\rm J}^{-1}{\rm s}^{-1}]$ because Φ is dimensionless, h [J] is the conjugate field, and the integral $\delta\langle\Phi\rangle = \int dt'\, R\, h$ supplies an extra [s] in the measure. The **macroeconomic exergy-bath temperature** $T_{\rm bath}$ is an effective temperature-like scale capturing stochastic fluctuations in stranded-energy availability. We define this thermal energy proportionally to the standard deviation of the exergetic chemical potential $\mu(t)$ over a macroscopic window $\tau_{\rm macro}$:

$$k_B T_{\rm bath} \equiv c_m \sqrt{\langle(\mu(t) - \langle\mu\rangle)^2\rangle_{\tau_{\rm macro}}}\,. \tag{78}$$

As miners operate at margins dictated by this stochastic availability, the temporal variance of $\mu(t)$ acts as the kinetic driver. Dominated by **active noise** $\zeta(t)$ from macroeconomic shocks, time-translation invariance breaks ($X < 1$). We define the **effective activity temperature** $T_{\rm act} \equiv T_{\rm bath}/X$ and introduce the corresponding active diffusion scale

$$D_{\rm act} \equiv \Lambda_0 k_B\,(T_{\rm act} - T_{\rm bath})\,, \tag{79}$$

which has units of ${\rm s}^{-1}$, consistent with the noise correlation $\langle\zeta(t)\zeta(t')\rangle$. The noise correlation can then be written as

$$\langle\zeta(t)\zeta(t')\rangle = 2\Lambda_0 k_B T_{\rm act}\,\delta(t-t')\,, \tag{80}$$

so that

$$X = \left(1 + \frac{D_{\rm act}}{k_B \Lambda_0 T_{\rm bath}}\right)^{-1}. \tag{81}$$

The network's elevated temperature directly reflects the mechanical work expended to maintain the UTXO Markov blanket.

To bridge these microscopic stochastic trajectories to the macroscopic ensemble observables measured empirically in Sec. VII C, we map the generalized Langevin dynamics onto the corresponding Fokker–Planck equation. Let $\rho(\Phi, t)$ represent the probability density of the network residing in macroscopic state Φ at time t. The temporal evolution of this probability density is governed by:

$$\frac{\partial \rho(\Phi,t)}{\partial t} = -\frac{\partial}{\partial \Phi}\left[\left(-\Lambda_0 \frac{\partial \mathcal{U}}{\partial \Phi} + \mathcal{A}(t)\right)\rho\right] + \Lambda_0 k_B T_{\rm act} \frac{\partial^2 \rho}{\partial \Phi^2}, \tag{82}$$

where the drift vector is dynamically modulated by the anticipatory force $\mathcal{A}(t)$, and the diffusion coefficient is $\Lambda_0 k_B T_{\rm act}$ [s^{-1}]. The probability density $\rho(\Phi, t)$ is normalized on the physical domain $\Phi \in [0, \infty)$ (since negative hashrate is unphysical):

$$\int_0^\infty \rho(\Phi, t)\, d\Phi = 1\,. \tag{83}$$

Consistent with the reflective boundary introduced below Eq. (69), we impose a zero-flux (reflecting) condition at $\Phi = 0$:

$$\left[v(\Phi,t)\,\rho - \frac{1}{2}\frac{\partial}{\partial \Phi}\big(\mathscr{B}(\Phi,t)\,\rho\big)\right]_{\Phi=0} = 0\,, \tag{84}$$

where $v(\Phi, t)$ [s^{-1}] is the drift coefficient and $\mathscr{B}(\Phi, t)$ [s^{-1}] is the diffusion coefficient of the corresponding Fokker–Planck equation (Appendix VIII C), consistent with Φ being a strictly dimensionless order parameter. This zero–probability-current condition ensures total probability conservation, supplemented by the natural decay condition $\rho(\Phi \to \infty, t) \to 0$.

During a thermodynamic quench (the Halving), the sudden extraction of the subsidy creates a transient divergence in the drift term. Before the DAA thermostat can dynamically update $\bar{E}_{\rm eff}$ to restore the potential well, the restoring force $-\partial\mathcal{U}/\partial\Phi$ weakens. Equation (82) dictates that this weakened drift, competing against constant active diffusion, causes the probability distribution $\rho(\Phi, t)$ to flatten and widen. This FPE formalization provides the exact theoretical mechanism for the transient spike in active-noise variance (distribution broadening) that we empirically detect in the network's block-discovery times.

For extreme non-Gaussian macroeconomic shocks, the active noise generalizes to heavy-tailed Lévy ν-stable noise, which would convert Eq. (82) into a fractional Fokker–Planck equation. However, our empirical metrology in Sec. VII C isolates discrete phase windows where the variance remains functionally finite, permitting standard statistical validation of the anticipatory and quench dynamics.

C. Thermodynamic Halving Quench and Thermostat (DAA)

The Halving imposes an abrupt phase transition on this tensioned system. The subsidy collapse renders the state $\Phi(t_H)$ metastable for inefficient operators. Due to sunk CapEx (thermodynamic inertia), miners unplug only when OpEx exceeds profitability. The resulting hashrate drop against a rigid Difficulty $\mathcal{D}$ diverges mean block wait time $\langle \Delta t \rangle$.

The system escapes this transient entropic spiral via the Difficulty Adjustment Algorithm (DAA). As established in Sec. III D, the DAA acts as an extended dynamic variable. During a quench, it functions in a manner reminiscent of a Nosé–Hoover thermostat [35], introducing a deterministic feedback force that damps runaway variance. Unlike the Nosé–Hoover scheme, which is time-reversible and conserves a generalized Hamiltonian, the DAA is a discrete, non-invertible feedback controller; the analogy is therefore restricted to their shared role as variance-regulating dynamical extensions. In the continuum limit, the thermodynamic friction responds to deviations from the target crystallization timescale:

$$\frac{d\mathcal{D}(t)}{dt} = \nu_{\rm DAA}\, \mathcal{D}(t) \left(\frac{\tau_B}{\langle \Delta t \rangle_{\rm epoch}} - 1\right), \tag{85}$$

where $\nu_{\rm DAA}$ acts as the protocol's fundamental feedback coupling rate (s^{-1}). Driven by this feedback loop, the system undergoes damped oscillations before restoring its dynamic mass toward a new metastable equilibrium.

Empirical Consistency Check: Sensitivity Analysis of Anticipation and Quench

We test $\mathcal{A}(t)$ [Eq. (74)] against inter-block times Δt across all four historical Halvings. The Anticipation–Quench boundary is fixed by sweeping the lookback horizon $k = 1$–14 epochs ($\lesssim 6.5$ months) and selecting the k that jointly maximizes the Fisher F-statistic and the Levene test statistic, eliminating *ad hoc* phase boundaries.

Halving Year	Opt. Horizon (k Epochs)	Quench Blocks	Anticipation Var(Δt)	Quench Var(Δt)	Statistical Tests %Δ Var	F-ratio	Fisher (p)
2012	14	1680	99.55	125.29	+25.86%	1.259	< 0.001
2016	14	1344	90.65	100.63	+11.00%	1.110	0.0069
2020	3	1008	90.21	139.56	+54.71%	1.547	< 0.0001
2024	1	672	87.33	97.18	+11.28%	1.113	0.0853

Table II: **Thermodynamic quench metrology (sensitivity analysis).** Comparison of block-time variance between the optimized Anticipation Phase (representing force $\mathcal{A}(t)$) and the transient Quench Phase. The shift from long ($k = 14$) to compressed ($k = 1, 3$) horizons reflects hardware-deployment maturation. Variance in s^2; F-ratio is the standard Fisher statistic for variance homogeneity.

This multi-halving analysis is *consistent with* the hypothesis that the ledger responds to shocks not merely through production drops, but through a non-equilibrium regime shaped by both forward-looking exergy injection and rapid thermostat intervention. We stress that the anticipatory drift $\mathcal{A}(t)$ is a phenomenological input:

the present data do not uniquely determine its functional form, and the empirical comparison should therefore be interpreted as a consistency check rather than a parameter-free confirmation of the model.

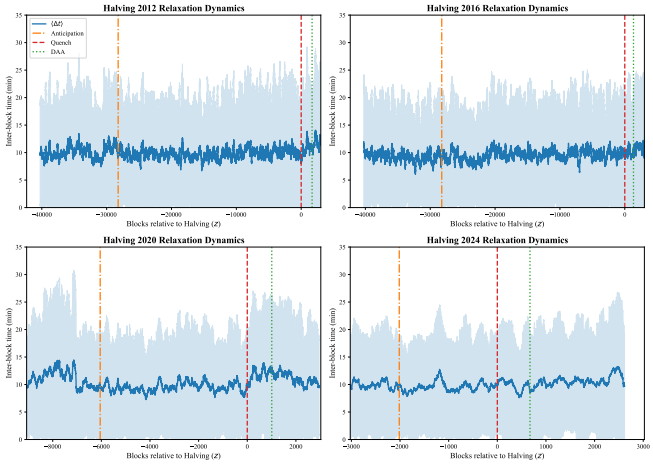

Figure 11: **Empirical relaxation dynamics of the thermodynamic quench.** Moving average of inter-block time $\langle \Delta t \rangle$ (solid blue) and thermal noise variance $\pm 1\sigma$ (shaded) relative to the Halving (dashed red). The phase boundaries reflect the optimal anticipation horizon isolated by our statistical sensitivity analysis.

As demonstrated in Fig. 11 and Table II, the optimized phase integration reveals a profound evolution in network thermodynamics corresponding to industrial maturity:

1. **Compression of anticipation horizons.** The optimal pre-Halving lookback contracted from 14 epochs (2012, 2016) to 3 epochs (2020) and 1 epoch (2024), consistent with increasing industrial maturity and faster capital deployment cycles.

2. **Quench divergence (2020).** The 2020 Halving produced the largest variance spike (+54.71%, $F = 1.547$, $p < 10^{-20}$), providing the cleanest empirical signature of drift-dominated to diffusion-dominated dynamics.

3. **Frustrated divergence (2024).** The 2024 quench variance increase (+11.28%, $p = 0.085$) was rapidly truncated by the DAA within 672 blocks, illustrating the thermostat's damping power in the industrialized regime.

4. **Fokker–Planck shape corroboration.** In 2012, 2020, and 2024, the quench-phase kurtosis drops systematically (e.g., $6.34 \to 5.11$ in 2020), confirming the predicted transition from leptokurtic (drift-dominated) to platykurtic (diffusion-dominated) distributions. The anomalous kurtosis increase in 2016 ($5.99 \to 7.10$) is consistent with a heavier-tailed effective noise source, plausibly associated with the concurrent 16 nm ASIC deployment cycle, but does not by itself establish a Lévy-stable regime or uniquely identify the underlying mechanism.

D. Testable Predictions and Operational Diagnostics

Because the present framework is phenomenological, its value depends on whether it produces operationally testable signatures. The following diagnostics summarize the principal predictions of the model:

1. **Latency-controlled effective temperature.** The definition $k_B T_{\text{eff}} \equiv (\epsilon_0^2/\bar{E}_{\text{eff}})\,(\tau_L/\tau_B)$ [Eq. (8)], where ϵ_0 [J] is the single-hash energy cost (the minimal work expended per trial nonce), $\bar{E}_{\text{eff}}$ [J] is the mean effective energy per block, and τ_L/τ_B is the dimensionless latency ratio, yields $[k_B T_{\text{eff}}] = \text{J}^2/\text{J} \cdot 1 = \text{J}$ as required. The ratio $\epsilon_0^2/\bar{E}_{\text{eff}}$ may be understood as the variance-to-mean energy ratio of the mining Poisson process, providing a physical interpretation analogous to thermal fluctuation strength. At fixed $\bar{E}_{\text{eff}}$ and τ_B, increases in propagation latency τ_L should systematically correlate with elevated orphan risk and reduced local ordering stability.

2. **Protocol-dependent upper bound on block volume.** The bound $V_{\text{B,crit}} \approx C_{\text{eff}}\,(\tau_B/\Gamma_c - \tau_0)$ [Eq. (27)], where C_{eff} [byte s^{-1}] is the effective network validation throughput, Γ_c [1] is a dimensionless safety factor ($\Gamma_c > 1$, representing the fraction of the block interval consumed by propagation and validation), and τ_0 [s] is a protocol-overhead latency floor, predicts that any protocol modification that increases validation friction γ or effective hop diameter D_{hop} should reduce the admissible block volume before large-scale desynchronization appears.

3. **Depth-dependent reorganization suppression.** In the ordered regime, the model predicts $\mathcal{P}_{\text{reorg}}(z) \sim e^{-z/\xi}$ with $\xi^{-1} = \ln(p/q)$ [Eq. (47)], implying that empirical estimates of reorganization risk in the ordered regime should collapse onto a large-depth exponential law when conditioned on the honest/attacker hashrate ratio. This is consistent with the path-weight framework: identifying $\beta_{\text{eff}}\,\langle E_{\text{eff}} \rangle = \ln(p/q)$ per block yields $\mathcal{P}_{\text{reorg}}(z) \sim \exp\!\big[-z\,\beta_{\text{eff}}\,\langle E_{\text{eff}} \rangle\big]$, reproducing the mass-gap scaling with $\xi^{-1} = \beta_{\text{eff}}\,\langle E_{\text{eff}} \rangle = \ln(p/q)$.

4. **Quench-induced excess variance.** Deterministic subsidy reductions should generate transient increases in block-time variance, with amplitude modulated by the anticipation horizon and by the damping action of the DAA.

5. **Compression of anticipation horizons with industrial maturation.** If the anticipatory force $\mathcal{A}(t)$ captures real forward-looking deployment dynamics, then the optimal pre-quench horizon extracted from historical data should shorten as min-

ing supply chains become more industrialized and capital deployment becomes more responsive.

These predictions are deliberately stated in operational form so that they may be confronted with simulation studies, historical chain data, or controlled stress tests on experimental Nakamoto-style networks.

VIII. CONCLUSION: THE PHYSICS OF NAKAMOTO DISTRIBUTED CONSENSUS

We model the Nakamoto blockchain as a **dissipative structure** maintained far from equilibrium. Mapping the ledger to a one-dimensional network reveals consensus as an **effective ordering transition**, where the exergy-driven hashrate order parameter spontaneously selects a unique chain history, breaking permutation symmetry among valid block sequences. The transition shares features of both continuous (Ginzburg–Landau) and first-order (nucleation-barrier) character depending on the observable considered.

The present theory is intentionally phenomenological. It does not derive the path measure, effective potential, or stochastic coefficients from a microscopic agent-based model of miner behavior, nor does it establish an exact universality-class correspondence with standard equilibrium or non-equilibrium systems. Its main claim is narrower: protocol observables admit a dimensionally consistent coarse-grained organization in which consensus stability, latency-induced disordering, and subsidy-driven quenches can be expressed through standard statistical-physics structures and confronted with data.

A. Model Summary

Our analysis yields five key physical perspectives:

1. **Thermal Time and Information Temperature:** Consensus time emerges as a coarse-grained quantity associated with irreversible dissipation, formalizing an effective network temperature (T_{eff}) that bridges information physics with thermodynamic stability.

2. **Thermodynamic Depth:** Ledger immutability is probabilistic, governed by a correlation length ξ decaying exponentially with cumulative work—resolving double spending as suppression of thermal fluctuations.

3. **Crystallization and Spatial Limit:** Mining is a phase boundary transitioning information from a "liquid" to a "crystalline" state. This stability is conditioned by a dimensionless ratio Γ that imposes an effective limit on block size to avoid thermal percolation.

4. **Dynamic Stability:** Network resilience to macroeconomic shocks ("Halving") is explained by first-order quench dynamics, effectively damped by a protocol thermostat (DAA) that provides a restorative force toward the new equilibrium.

5. **Maxwell's Demon and Informational Erasure:** As a macroscopic demon, the network sorts the high-entropy mempool into the ordered UTXO set. Reducing Shannon uncertainty across this blanket requires irreversible updates, incurring a Landauer-type dissipation bound. This remains a phenomenological analogy, not a literal microscopic mechanism.

B. Thermodynamic Inertia and a Phenomenological Mass Analogy

Our framework suggests a phenomenological ledger inertia, generating a mass-like resistance to history reversal through cumulative Proof-of-Work dissipation. We define this **Global Mass** of history, $M_{\text{global}}^{(\text{phen})} \equiv M_{\text{eff}}^{(\text{phen})}$, by normalizing the Consensus Action $\mathcal{S}_{\text{PoW}}$ [Eq. (43)] by the hardware's embodied specific exergy e_{mfg} [$\text{J}\,\text{kg}^{-1}$] [Sec. III B]:

$$M_{\text{eff}}^{(\text{phen})} \equiv \frac{1}{e_{\text{mfg}}\,\tau_B} \int_{t_1}^{t_2} \bar{E}_{\text{eff}}(t)\,dt = \frac{\mathcal{S}_{\text{PoW}}(t_1,t_2)}{e_{\text{mfg}}\,\tau_B}\,. \quad (86)$$

This global inertia corresponds to the integrated volume of the potential well since genesis.

Conversely, resistance to *local* perturbations (e.g., chain-tip reorganizations) is dictated by the **Marginal Mass** of consensus. By analogy with massive field theories, we isolate this restoring force via the local curvature of the Landau potential at the equilibrium order parameter Φ_0:

$$\kappa_{\text{marginal}} \equiv \frac{1}{2}\frac{\partial^2 \mathcal{U}}{\partial \Phi^2}\bigg|_{\Phi_0} = -2\,\alpha_{\text{h}}(T_{\text{eff}})\,. \quad (87)$$

In the ordered phase ($\alpha_{\text{h}} < 0$), this energetic stiffness κ_{marginal} is strictly positive ([J]).

We relate this stiffness to the topological mass gap $\mathfrak{m} = \xi^{-1}$ [Sec. IV B] via a quasi-Boltzmann ansatz. Assuming the canonical chain's per-block statistical advantage (p/q) scales as $\exp(\kappa_{\text{marginal}}/k_B T_{\text{eff}})$, taking the logarithm yields:

$$\mathfrak{m} = \xi^{-1} = \ln\left(\frac{p}{q}\right) = \frac{\kappa_{\text{marginal}}}{k_B T_{\text{eff}}}\,. \quad (88)$$

This mean-field identification strictly equates the dimensionless mass gap to the ratio of local restoring force to effective thermal noise. The correlation length $\xi = a_0/\mathfrak{m}$ remains a dimensionless block count.

In the near-critical regime ($p/q \approx 1+\epsilon$), leading-order expansion yields:

$$\frac{p-q}{q} \approx \frac{\kappa_{\text{marginal}}}{k_B T_{\text{eff}}}\,. \quad (89)$$

The honest network's relative hashrate advantage thus functions as an effective **thermodynamic signal-to-noise ratio**. Modulated by the difficulty target, this marginal mass dictates the macroscopic correlation length ξ, protecting the active tip against thermal evaporation. Consequently, while Global Mass anchors irreversible history, Marginal Mass provides dynamic resistance to temporal reversibility.

Under Landauer's Principle [4], this immutability stems from continuous **uncertainty erasure**. The massive phenomenological inertia $M_{\text{eff}}^{(\text{phen})}$ represents the macroscopic remanence of countless physically destroyed, invalid microstates. This irreversible informational erasure creates an ever-deepening free-energy well, rendering history reversal a thermodynamically implausible fluctuation against the accumulated dissipation barrier $\Delta\mathcal{F}$.

C. Asymptotic Physical Limits and Broad Applicability

As a rough consistency check, one may estimate the asymptotic floor of h_N by combining the Landauer limit at $T_{\text{sub}} = 300\,\text{K}$ ($k_B T_{\text{sub}} \ln 2 \approx 2.87 \times 10^{-21}\,\text{J}$ per irreversible bit operation, with $\sim 10^5$ such operations per SHA-256 hash) with femtosecond-scale switching times, yielding $h_{N,\text{limit}} \sim 10^{-31}\,\text{J}\cdot\text{s}$—within a few orders of magnitude of Planck's constant h. However, material constraints intervene well before this limit: at the current computational mass efficiency $\eta_M \approx 1.5 \times 10^{-13}\,\text{kg}\cdot\text{s}$, exploiting the full solar irradiance ($\sim 10^{17}\,\text{W}$) would demand a hardware mass exceeding $10^{19}\,\text{kg}$, indicating that material, rather than purely energetic, bounds will dominate in practice.

The protocol may thus be interpreted as anchoring cryptographic security in physical resource constraints and irreversible dissipation. By formalizing consensus as a non-equilibrium phase transition, this framework yields falsifiable limits, including an absolute bound on block volume ($V_{\text{B,crit}}$) to avert topological collapse and predicted variance divergences during exergetic quenches.

It would be interesting to explore whether analogous dissipative ordering mechanisms appear in other systems sustained far from equilibrium, such as neural synchronization or cellular proofreading, though such extensions lie beyond the scope of the present work. Notably, this framework mathematically parallels recent observations linking the thermodynamics of Nakamoto consensus to cellular biology, where ribosomes utilize kinetic proofreading as a biological "proof-of-work" to reduce thermal noise and assemble proteins accurately from RNA code.

More modestly, the present formalization places the distributed ledger at an instructive intersection of thermodynamics, information theory, and decentralized computation. By coupling cryptographic asymmetry [36] with Shannon information [7] and irreversible dissipation, the framework suggests how large-scale consensus can be analyzed as an active ordering process subject to energetic, informational, and material constraints. In that sense, Nakamoto consensus may serve as a useful laboratory for studying how decentralized architectures resist entropic degradation under sustained non-equilibrium forcing.

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Author contributions. P.R. conceived the theoretical framework, performed the analytical calculations, and wrote the manuscript. J.Y. contributed to the empirical analysis and edited the manuscript.

Appendix A: Summary of Physical Variables. Principal variables and dimensional assignments used in the phenomenological model. Assignments are chosen to maintain internal consistency across the Landau description, global-mode stochastic dynamics, and action-like path-weight constructions. The notation distinguishes T_{sub} (hardware temperature), T_{act} (stochastic activity temperature), and T_{bath} (macroeconomic exergy-bath temperature). Variables are listed in order of first appearance within each category. *Key*: E =energy, T =time, L =length, Θ =temperature, M =mass, Info =information. Epistemic status: **(i)** directly protocol-level quantity or dimensional-analysis result, **(ii)** effective phenomenological coefficient or state variable, **(iii)** interpretive quantity introduced by physical analogy (see main text, Sec. I).

Sym.	Description	Sec.	Dim.	SI	Ep.
	I. Temporal and kinematic variables				
t	External physical time	I	[T]	s	(i)
τ_B	Target block interval	II.C	[T]	s	(i)
τ_L	Network propagation latency	II.C	[T]	s	(i)
τ_{hop}	Single-hop latency	III.A	[T]	s	(i)
τ_0	Geometric delay floor	III.A	[T]	s	(i)
τ_H	Subsidy Halving interval	VII.A	[T]	s	(i)
τ_{hash}	SHA-256 transition time	III.E	[T]	s	(i)
τ_{plan}	Anticipation planning horizon	VII.B	[T]	s	(ii)
τ_{macro}	Macroscopic averaging window	VII.B	[T]	s	(ii)
Δt	Inter-block wait time	VII.C	[T]	s	(i)
N	Ledger time (block height)	II.A	Count [1]	blocks	(i)
z	Reorganization depth	IV.A	Count [1]	blocks	(i)
$H(t)$	Global network hashrate	III.D	$[\mathrm{T}^{-1}]$	s^{-1}	(i)
H_{eq}	Adiabatic equil. hashrate	VII.B	$[\mathrm{T}^{-1}]$	s^{-1}	(ii)
r_{tech}	Technology innovation rate	VII.A	$[\mathrm{T}^{-1}]$	s^{-1}	(ii)
λ	Temporal discount rate	VII.B	$[\mathrm{T}^{-1}]$	s^{-1}	(ii)
ν_{DAA}	DAA feedback coupling rate	VII.C	$[\mathrm{T}^{-1}]$	s^{-1}	(i)
$v(\Phi,t)$	FPE drift coefficient	VII.B	$[\mathrm{T}^{-1}]$	s^{-1}	(ii)
$\mathscr{D}(\Phi,t)$	FPE diffusion coefficient	VII.B	$[\mathrm{T}^{-1}]$	s^{-1}	(ii)
ν_H	Exergetic growth exponent	VII.A	[1]	–	(ii)
$\phi(x,t)$	Local GL order parameter	III	[1]	–	(ii)
$\Phi(t)$	Global exergy order parameter	VII.B	[1]	–	(ii)
$h(t)$	Conjugate source (FDR)	VII.B	E	J	(iii)
Λ_0	Anticipation susceptibility	VII	$[\mathrm{E}^{-1}\mathrm{T}^{-1}]$	$\mathrm{J}^{-1}\mathrm{s}^{-1}$	(ii)
g_{ant}	Anticipation coupling	VII	[1]	–	(ii)
λ	Anticipation decay rate	VII	$[\mathrm{T}^{-1}]$	s^{-1}	(ii)
	II. Thermodynamic and energetic variables				
k_B	Boltzmann constant	II.C	E/Θ	$\mathrm{J\,K}^{-1}$	(i)
T_{eff}	Effective information temperature	II.C	Θ	K	(ii)
T_c	Critical effective temperature	II.D	Θ	K	(ii)
T_{sub}	Substrate (environment) temp.	V.B	Θ	K	(i)
T_{bath}	Exergy-bath temperature	VII.B	Θ	K	(iii)
T_{act}	Effective activity temperature	VII.B	Θ	K	(ii)
T_{KT}	Kosterlitz–Thouless temp.	VI.C	Θ	K	(iii)
β_{eff}	Effective inverse energy scale	II.A	E^{-1}	J^{-1}	(ii)
$\bar{E}_{\mathrm{eff}}$	Mean energy per block	II.C	E	J	(i)
$E_{\mathrm{eff}}(B_i)$	Effective block energy	II.A	E	J	(ii)
ϵ_0	Microscopic energy threshold	II.C	E	J	(i)
$\mathcal{E}_{\mathrm{shot}}$	Poissonian shot-noise energy scale	II.D	E	J	(ii)
$\Sigma[\mathcal{C}]$	Cumulative dissipative path functional	II.A	E	J	(ii)
$\mathcal{F}$	GL free energy (total)	II.C	E	J	(ii)
$f(\phi)$	Local free-energy density (1D)	III.B	E/L	$\mathrm{J\,m}^{-1}$	(ii)
$a,\ b$	Landau GL coefficients	III.B	E/L	$\mathrm{J\,m}^{-1}$	(ii)
κ	Gradient stiffness (1D)	III.B	E·L	J m	(ii)
χ_E	Free-energy–temp. susceptibility	III.B	$\mathrm{E}/(\mathrm{L}\Theta)$	$\mathrm{J\,m}^{-1}\mathrm{K}^{-1}$	(ii)
$\mathcal{E}_{\mathrm{coupling}}$	Coupling free-energy density	III.B	E/L	$\mathrm{J\,m}^{-1}$	(ii)
c_0	Saturation structural constant	III.B	[1]	–	(ii)
e_{mfg}	Embodied ASIC exergy density	III.B	E/M	$\mathrm{J\,kg}^{-1}$	(i)
$e_{\mathrm{hash}}(t)$	Energy per hash	III.D	E	J	(i)
$P_{\mathrm{net}}(t)$	Instantaneous network power	III.D	Power	W	(i)
$P_{\mathrm{eq}}(t)$	Equilibrium network power	III.D	Power	W	(ii)
$P_{\mathrm{in}}(t)$	Thermodynamic power influx	VII.A	Power	W	(ii)
σ_{obs}	Entropy production (window)	III.D	E/Θ	$\mathrm{J\,K}^{-1}$	(ii)
S_{th}	Thermodynamic entropy of history ensemble	II.C	E/Θ	$\mathrm{J\,K}^{-1}$	(ii)
$\dot{S}_{\mathrm{irrev}}$	Irrev. entropy production rate	VII	$\mathrm{E}\Theta^{-1}\mathrm{T}^{-1}$	$\mathrm{J\,K}^{-1}\mathrm{s}^{-1}$	(ii)
$\dot{Q}_{\mathrm{diss}}$	Heat dissipation rate	V.C	Power	W	(i)
$\Delta\mathcal{F}(z)$	Reorg. free-energy barrier	IV.C	E	J	(ii)
$\bar{J}$	XY-model coupling constant	VI.A	E	J	(iii)
E_{vortex}	Vortex excitation energy	VI.C	E	J	(iii)
S_{vortex}	Vortex configurational entropy	VI.C	E/Θ	$\mathrm{J\,K}^{-1}$	(iii)
$\mu(t)$	Effective chemical-potential-like exergy functional	VII.A	E	J	(ii)
$\mu_{\mathrm{exergy}}(t)$	Network exergy reservoir	VII.A	E	J	(ii)
$\mu_{\mathrm{subsidy}}(t)$	Block-subsidy exergy contribution	VII.A	E	J	(i)
μ_{max}	Maximum accessible exergy ceiling	VII.A	E	J	(ii)
$\Delta\mu$	Discrete subsidy-drop magnitude	VII.A	E	J	(i)
$\mathcal{U}(\Phi)$	Effective Landau potential	VII.B	E	J	(ii)
$\alpha_h,\ b_h$	Landau expansion coefficients	VII.B	E	J	(ii)
$\mathcal{A}(t)$	Anticipatory drift term	VII.B	T^{-1}	s^{-1}	(ii)
Λ_0	Kinetic mobility coefficient	VII.B, App. C	$\mathrm{E}^{-1}\mathrm{T}^{-1}$	$\mathrm{J}^{-1}\mathrm{s}^{-1}$	(ii)
g_{ant}	Anticipation coupling constant	VII.B	1	1	(ii)
λ	Temporal discount rate	VII.B	T^{-1}	s^{-1}	(ii)
τ_{plan}	Anticipation horizon	VII.B	T	s	(ii)
$\kappa_{\mathrm{marginal}}$	Potential curvature scale	VIII.B	E	J	(ii)
	III. Order-parameter and stochastic variables				
$\Phi(t)$	Global order parameter	II.D	[1]	–	(ii)
$\phi(x,t)$	Local order parameter (1D field)	III.B	[1]	–	(ii)
Φ_0	Equilibrium order-param. value	III.B	[1]	–	(ii)
$\rho(\Phi,t)$	Probability density over Φ	VII.B	[1]	–	(ii)
Λ_0	Order-parameter mobility	VII.B	$\mathrm{E}^{-1}\mathrm{T}^{-1}$	$\mathrm{J}^{-1}\mathrm{s}^{-1}$	(ii)
$\zeta(t)$	Active stochastic noise	VII.B	T^{-1}	s^{-1}	(ii)
D_{act}	Active diffusion scale	VII.B	T^{-1}	s^{-1}	(ii)
X	Fluctuation–dissipation ratio	VII.B	[1]	–	(ii)
c_m	Exergy-volatility scaling const.	VII.B	[1]	–	(ii)
g_{ant}	Anticipation coupling constant	VII.B	[1]	–	(ii)
	IV. Correlation and topological variables				
$\mathbf{B}$	Chain configuration (set)	II.A	[1]	–	(i)
B_i	Individual block state	II.A	[1]	–	(i)
Ω	State space of fork histories	II.A	Count [1]	–	(i)
$\vec{s}_i$	Local consensus spin vector	II.D	[1]	–	(iii)
θ_i	Consensus phase variable	VI.A	Phase [1]	rad	(iii)
$G(z)$	Two-point correlation function	IV.A	[1]	–	(ii)
a_0	Block lattice spacing	IV.A	Topo. dist.	block	(i)
$\mathfrak{m}$	Dimensionless mass gap	IV.A	[1]	–	(ii)
$\xi,\ \xi_{\parallel}$	Reorganization corr. length	IV.A	Topo. dist.	blocks	(ii)
$p,\ q$	Honest / attacker hash share	IV.B	[1]	–	(i)
$\mathcal{P}(z)$	Fork survival probability	IV.D	[1]	–	(ii)
δ	DP survival exponent	IV.D	[1]	–	(iii)
$\nu_{\parallel}$	DP parallel corr. exponent	IV.D	[1]	–	(iii)
	V. Network and information variables				
$\Gamma,\ \Gamma_c$	Latency-control ratio / critical value	II.D	[1]	–	(i)
$V_B,\ V_{\mathrm{B,crit}}$	Block size / critical size	III.A	Info	Bytes	(i)
C_{node}	Bottleneck link capacity	III.A	Info/T	$\mathrm{B\,s}^{-1}$	(i)
γ	Validation-time coefficient	III.A	T/Info	$\mathrm{s\,B}^{-1}$	(i)
D_{hop}	Network diameter (hops)	III.A	Count [1]	hops	(i)
C_{eff}	Effective global capacity	III.A	Info/T	$\mathrm{B\,s}^{-1}$	(i)
L_{net}	Network spatial diameter	III.B	[L]	m	(i)
$\mathcal{D}$	Protocol difficulty	III.D	[1]	–	(i)
$\mathcal{N}_H(\mathcal{D})$	Expected hashes per block	III.D	Count [1]	hashes	(i)
H_{bit}	Shannon entropy (bits)	V.B	Info	bits	(i)
N_{bit}	Cumul. irreversible bit ops	V.C	Count [1]	bits	(i)
$\mathcal{V}$	Blockchain historical volume	V.A	State set [1]	–	(i)
$\partial\mathcal{V}$	UTXO Markov blanket	V.A	State set [1]	–	(iii)
$\mathcal{B}_t$	Transaction set in block t	V.A	State set [1]	–	(i)
$\mathcal{N}_{\mathrm{nodes}}(r)$	Nodes within radius r	VI.C	Count [1]	–	(i)
L	Network topological diameter	VI.C	Topo. dist.	hops	(i)
$d_{\mathrm{eff}},\ d_s$	Effective / spectral dimension	VI.C	[1]	–	(iii)
	VI. Action and inertia variables				
$h_N(t)$	Nakamoto action per hash	III.E	Action	J s	(ii)
$\mathcal{S}_{\mathrm{PoW}}$	Accumulated consensus action	III.E	Action	J s	(ii)
η_M	Computational mass efficiency	III.B	M·T	kg s	(ii)
$M_{\mathrm{eff}}^{(\mathrm{phen})}$	Phenomenological inertia	VIII.B	M	kg	(iii)
M_{hardware}	Physical hardware mass	VIII.C	M	kg	(i)

Appendix B: Plan of the Article & Logical Flow. Compact summary of each section and subsection, indicating how the phenomenological framework progresses from foundational concepts to coarse-grained dynamics and empirical observations.

ID	Section / Subsection Name	Physical Summary / Theoretical Role
	I. Introduction: The Thermodynamics of Distributed Consensus	
	Introduces consensus as an effective symmetry-breaking process sustained by dissipation rather than purely logical ordering. The dimensionless global order parameter Φ, the absolute active noise scale $k_B T_{\mathrm{act}}$ [J], and the effective block cost $E_{\mathrm{eff}}(B_i)$ [J] [Eq. (7)] are defined to ensure strict macroscopic dimensional coherence across all subsequent non-equilibrium dynamics.	
I.	Introduction: consensus as dissipative symmetry breaking.	Clarifies the phenomenological status of the model at three epistemic levels (i)–(iii), distinguishes derived coarse-grained consequences from heuristic analogies, and explicitly limits claims relative to microscopic statistical mechanics.
	II. Network Model, Statistics and the Arrow of Time	
	Maps ledger evolution to a driven non-equilibrium system weighted by a cumulative effective ledger cost (or action-like energy functional).	
II.A	The Minimal Network Model and Consensus Hamiltonian	Defines the blockchain as a 1D growing network and introduces the dissipation-weighted path measure.
II.B	Time-Reversal Symmetry and Sybil Degeneracy	Interprets costless identity proliferation as a high-degeneracy, high-temperature regime.
II.C	Microscopic Substrate and the Nakamoto Action	Defines the action-based hardware anchor linking microscopic switching events to macroscopic consensus.
II.D	Information Temperature and Thermodynamic Regimes	Defines T_{eff} from the ratio of latency to block interval and identifies ordered/disordered regimes.
II.E	Spontaneous Symmetry Breaking (SSB)	Introduces the global order parameter and the onset of macroscopic alignment.
II.F	Emergence of Thermal Time and Fluctuation Theorem	Uses heuristic thermal-time scaling and fluctuation relations to motivate irreversible ledger time.
	III. Thermodynamic Potential and Symmetry Breaking	
	Constructs a coarse-grained Landau description of consensus stability and protocol-dependent limits.	
III.A	Effective Temperature, Latency, and Block Size Limit	Derives an effective thermal bound on block size from propagation, bandwidth, and validation friction.
III.B	The Order Parameter and Landau Free Energy	Introduces the 1D Ginzburg–Landau functional and identifies how latency controls ordering.
III.C	Phase-like Fluctuations and Consensus Rigidity	Describes local branch disagreements as phase-like fluctuations penalized by spatial stiffness.
III.D	Macroscopic Energy Scale, the DAA Thermostat	Interprets the DAA as a feedback mechanism stabilizing the effective energy scale.
III.E	The Consensus Action and Cumulative Barrier	Defines the macroscopic accumulation of work as a thermodynamic barrier against reorganization.
	IV. Correlation Length and Probabilistic Finality	
	Finality as exponential correlation decay with depth.	
IV.A	Correlation Function	Tip-to-history correlations.
IV.B	Mass Gap ("6-Block Rule")	Reorganization suppression $\leftrightarrow$ inverse correlation depth.
IV.C	False-History Nucleation	History rewriting as barrier activation.
IV.D	Forks as Absorbing-State Transitions	Orphans decay toward canonical absorbing state.
	V. Information Entropy and Markov Blanket	
	Explains how the protocol compresses history into a reduced state description under finite communication constraints.	
V.A	The Volume, the Markov Blanket, and RG Flow	Describes validation as an irreversible coarse-graining onto the active UTXO state.
V.B	Shannon vs. Thermodynamic Entropy	Separates informational uncertainty from dissipative cost at hardware and consensus scales.
V.C	The UTXO Set as a Physical Maxwell's Demon	Uses the demon analogy to interpret filtering and erasure costs phenomenologically.
V.D	Blanket Saturation and Channel Limit	Identifies synchronization loss when update volume approaches effective channel capacity.
	VI. Topological Stability of the Network Graph	
	Studies transverse consensus stability on the peer-to-peer graph using phase ordering and topological defects.	
VI.A	The Consensus Phase Field	Assigns node phases and coupling energies to represent spatial synchronization.
VI.B	Forks as Topological Defects (Vortices)	Maps unresolved network forks to XY-model vortices using principal-value phase unwrapping; the continuous-phase interpretation applies in the mempool regime where nodes maintain fractional commitment across tips (Sec. VI A).
VI.C	Effective Dimensionality and Kosterlitz–Thouless Bound	Uses a reduced-dimensional limit to describe partition-driven loss of coherence.
	VII. Non-Equilibrium Dynamics: Exergetic Expansion and the "Halving" Quench	
	Analyzes temporal response to deterministic reward shocks in a driven non-equilibrium setting.	
VII.A	The Exergy-Driven Potential	Models long-run hashrate growth through an exergy-limited secular trend.
VII.B	Non-Equilibrium Stochastic Dynamics and FDT Violation	Introduces a global-order-parameter Langevin model with anticipatory forcing and active noise.
VII.C	Thermodynamic Halving Quench and Thermostat (DAA)	Uses sensitivity analysis of Halving windows to characterize non-equilibrium response and damping.
VII.D	Testable Predictions and Operational Diagnostics	Summarizes the principal falsifiable consequences of the framework.
	VIII. Conclusion: The Physics of Distributed Consensus	
	Summarizes the framework, its limits, and its broader interpretive scope.	
VIII.A	Model Summary	Recaps the main thermodynamic and information-physics perspectives.
VIII.B	Thermodynamic Inertia and Phenomenological Mass	Introduces a cautious mass-like analogy for resistance to history reversal.
VIII.C	Asymptotic Physical Limits	Discusses energetic and material limits and possible extensions to other dissipative systems.

APPENDIX C: KRAMERS–MOYAL DERIVATION OF THE FOKKER–PLANCK EQUATION

We derive the Fokker–Planck equation from the Markovian Langevin dynamics of Eq. (70). Starting from the coarse-grained equation for the dimensionless order parameter Φ,

$$\frac{d\Phi}{dt} = -\Lambda_0 \frac{\partial \mathcal{U}}{\partial \Phi} + \mathcal{A}(t) + \zeta(t)\,, \tag{C1}$$

we define the deterministic drift velocity

$$v(\Phi, t) = -\Lambda_0 \frac{\partial \mathcal{U}}{\partial \Phi} + \mathcal{A}(t)\,, \tag{C2}$$

with Gaussian, delta-correlated active noise

$$\langle \zeta(t)\, \zeta(t') \rangle = 2\Lambda_0 k_B T_{\rm act}\, \delta(t - t')\,. \tag{C3}$$

Because the noise is additive, the Itô and Stratonovich conventions lead to the same Fokker–Planck equation; we adopt the Itô convention for definiteness.

The Kramers–Moyal expansion of the transition probability yields

$$\frac{\partial P(\Phi, t)}{\partial t} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \frac{\partial^n}{\partial \Phi^n} \left[D^{(n)}(\Phi, t)\, P(\Phi, t) \right], \tag{C4}$$

where the n-th Kramers–Moyal coefficient is $D^{(n)}(\Phi, t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \langle [\Delta\Phi]^n \rangle_{\Phi,t} / n!$. For Gaussian white noise only $n = 1, 2$ survive: $D^{(1)} = v(\Phi, t)$ (the drift) and $D^{(2)} = \Lambda_0\, k_B T_{\rm act}$ (the diffusion constant), with all $D^{(n\geq 3)} = 0$ by Pawula's theorem applied to the Gaussian closure. Substituting into Eq. (C4) yields the Fokker–Planck equation

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial \Phi} \left[v(\Phi, t)\, P \right] + \Lambda_0\, k_B T_{\rm act} \frac{\partial^2 P}{\partial \Phi^2}\,. \tag{C5}$$

Since Φ is dimensionless, $[d\Phi/dt] = \mathrm{s}^{-1}$, hence $[\zeta] = \mathrm{s}^{-1}$ and the correlator has dimensions s^{-2}. The effective diffusion constant $D_\Phi \equiv \Lambda_0\, k_B T_{\rm act}$ has $[D_\Phi] = \mathrm{J}^{-1}\mathrm{s}^{-1} \cdot \mathrm{J} = \mathrm{s}^{-1}$, and $[\delta(t-t')] = \mathrm{s}^{-1}$, so $[2D_\Phi\, \delta(t-t')] = \mathrm{s}^{-2}$, as required.

The first two Kramers–Moyal coefficients are

$$D^{(1)}(\Phi, t) = \lim_{\Delta t \to 0} \frac{\langle \Phi(t + \Delta t) - \Phi(t) \rangle}{\Delta t} = v(\Phi, t)\,, \tag{C4}$$

$$D^{(2)} = \lim_{\Delta t \to 0} \frac{\langle (\delta\Phi_{\rm stoch})^2 \rangle}{2\,\Delta t} = \Lambda_0\, k_B T_{\rm act}\,, \tag{C5}$$

Since the effective activity energy scale $k_B T_{\rm act}$ carries dimensions of $[\mathrm{J}]$ and the mobility Λ_0 is $[\mathrm{J}^{-1}\mathrm{s}^{-1}]$, we have $[D^{(1)}] = [D^{(2)}] = \mathrm{s}^{-1}$, ensuring every term in the Fokker–Planck equation is perfectly dimensionally consistent with the temporal evolution of the dimensionless order parameter Φ.

Pawula's theorem [37] guarantees that for a continuous Markov process all coefficients $D^{(n\geq 3)}$ vanish, ensuring $\rho(\Phi, t) \geq 0$. The evolution equation is therefore

$$\partial_t \rho = -\partial_\Phi \big[D^{(1)} \rho \big] + \partial_\Phi^2 \big[D^{(2)} \rho \big]\,. \tag{C6}$$

For time-independent coefficients the stationary solution takes Boltzmann–Gibbs form,

$$\rho_{\rm st}(\Phi) \propto \exp\Big[-U_{\rm eff}(\Phi) / k_B T_{\rm act} \Big], \quad U_{\rm eff}(\Phi) \equiv -\int^{\Phi} \frac{D^{(1)}(\Phi')}{D^{(2)}(\Phi')}\, d\Phi'\,, \tag{C7}$$

connecting the Langevin dynamics of Sec. VII D to the Landau–Ginzburg free-energy landscape of Sec. III and closing the logical loop of the phenomenological construction. A non-Markovian extension via Mori–Zwanzig projection is left for future work.

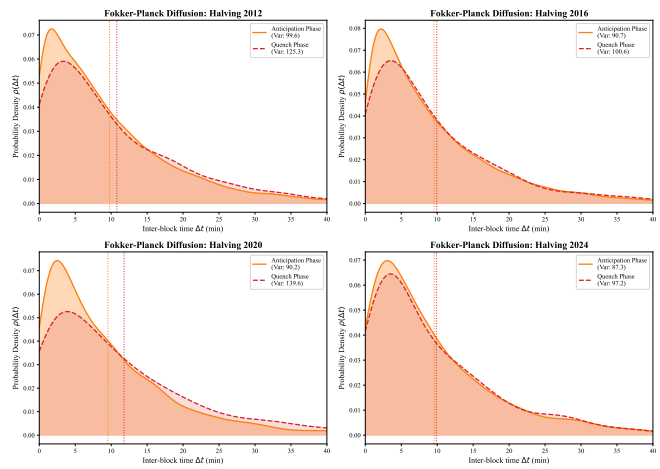

Figure 12: **Empirical corroboration of Fokker–Planck diffusion.** KDE of inter-block-time densities during the Anticipation (solid) and Quench (dashed) phases. The subsidy collapse weakens deterministic drift relative to active diffusion, broadening the distribution. The 2016 anomaly may indicate heavier-tailed noise beyond the Gaussian closure; quantitative Lévy discrimination is left for future work.

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